

Cloud Storage & Networking Basics Activity

Cloud Computing

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Activity 1 - Create and configure an S3 bucket

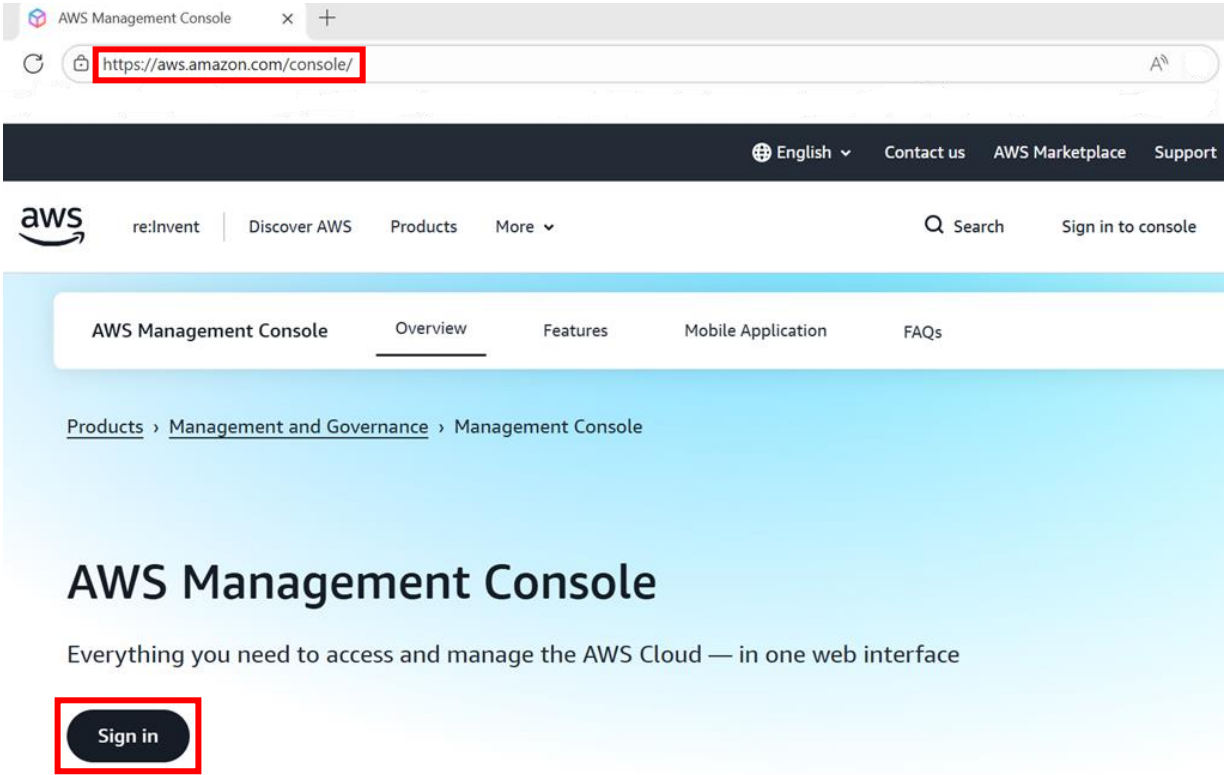
1.1 Prerequisite

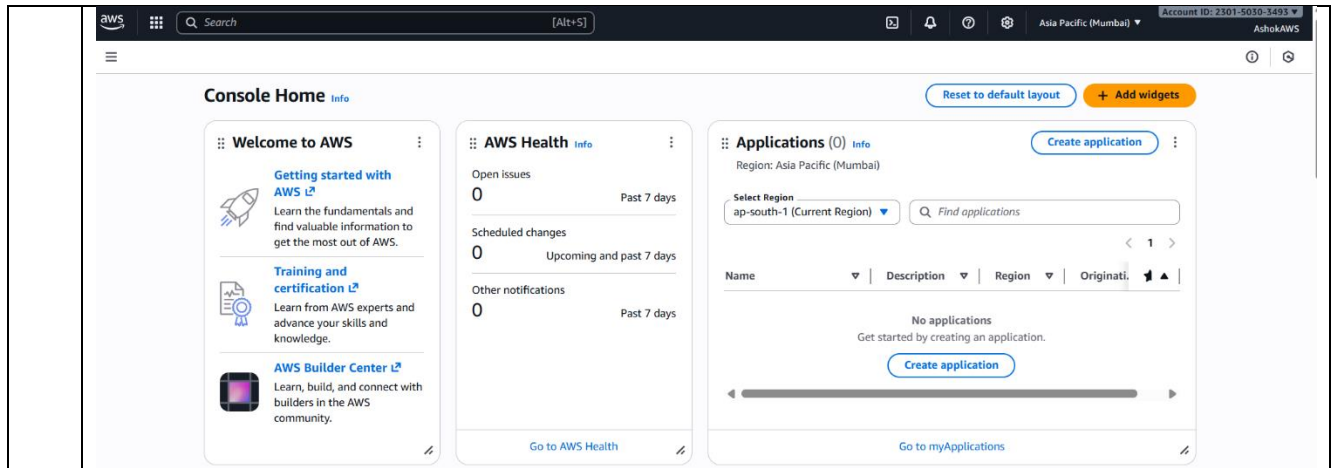
- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

1.2 Scenario

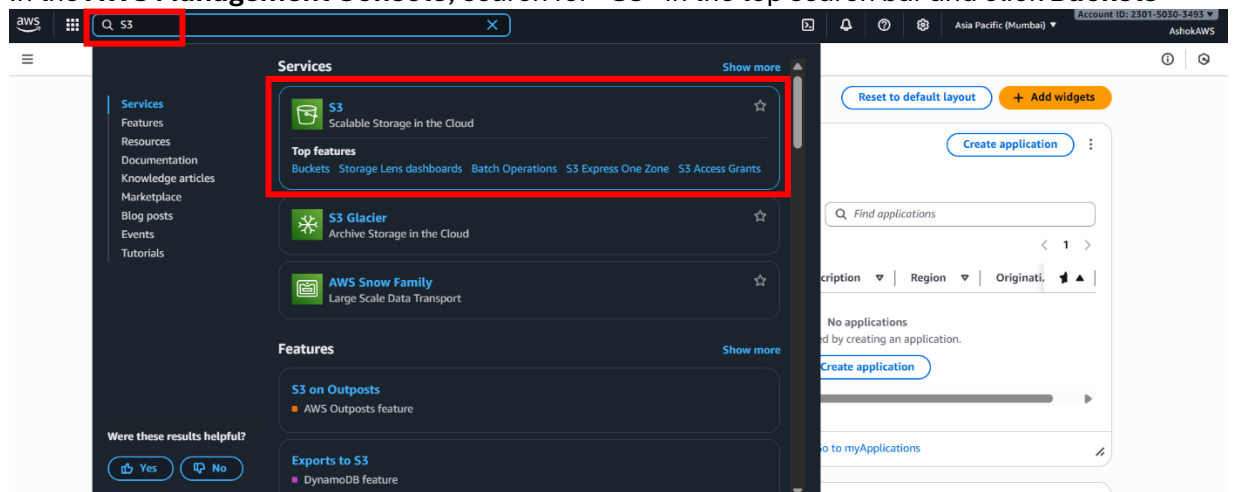
Create and configure an S3 bucket

1.3 Steps

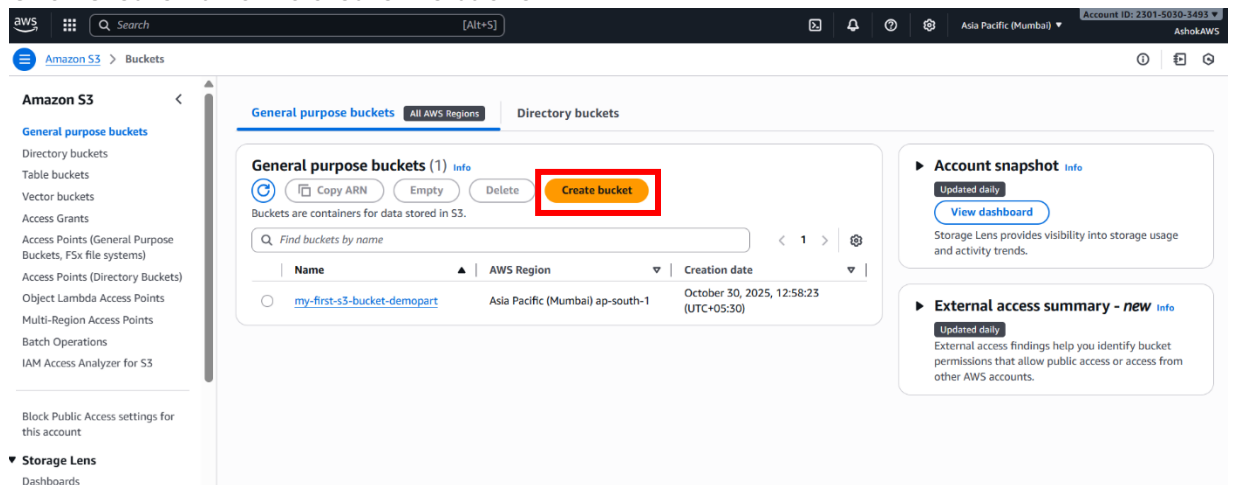
1	Log in to AWS Console • Visit https://aws.amazon.com/console • Sign in with your credentials  After Sign in, the following page will be displayed:
---	--



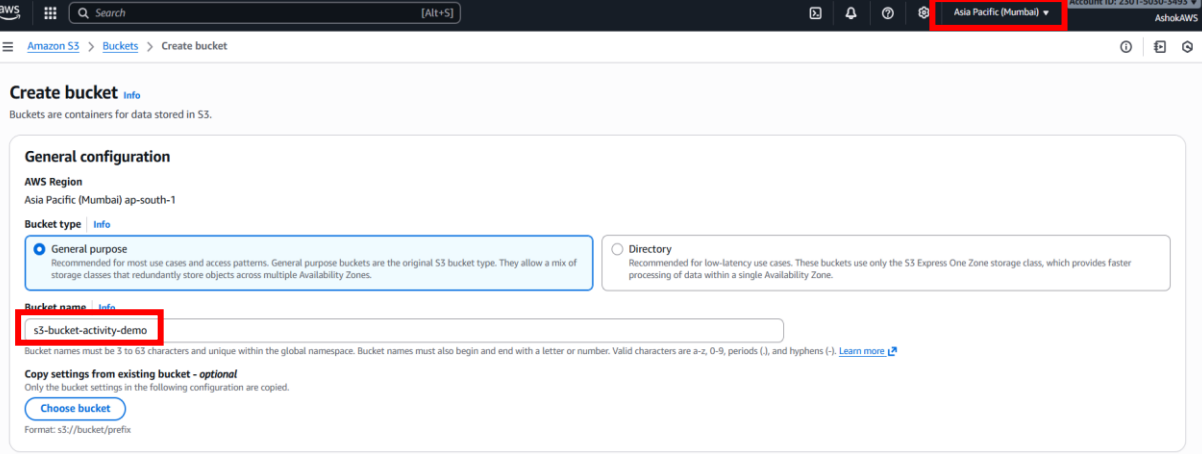
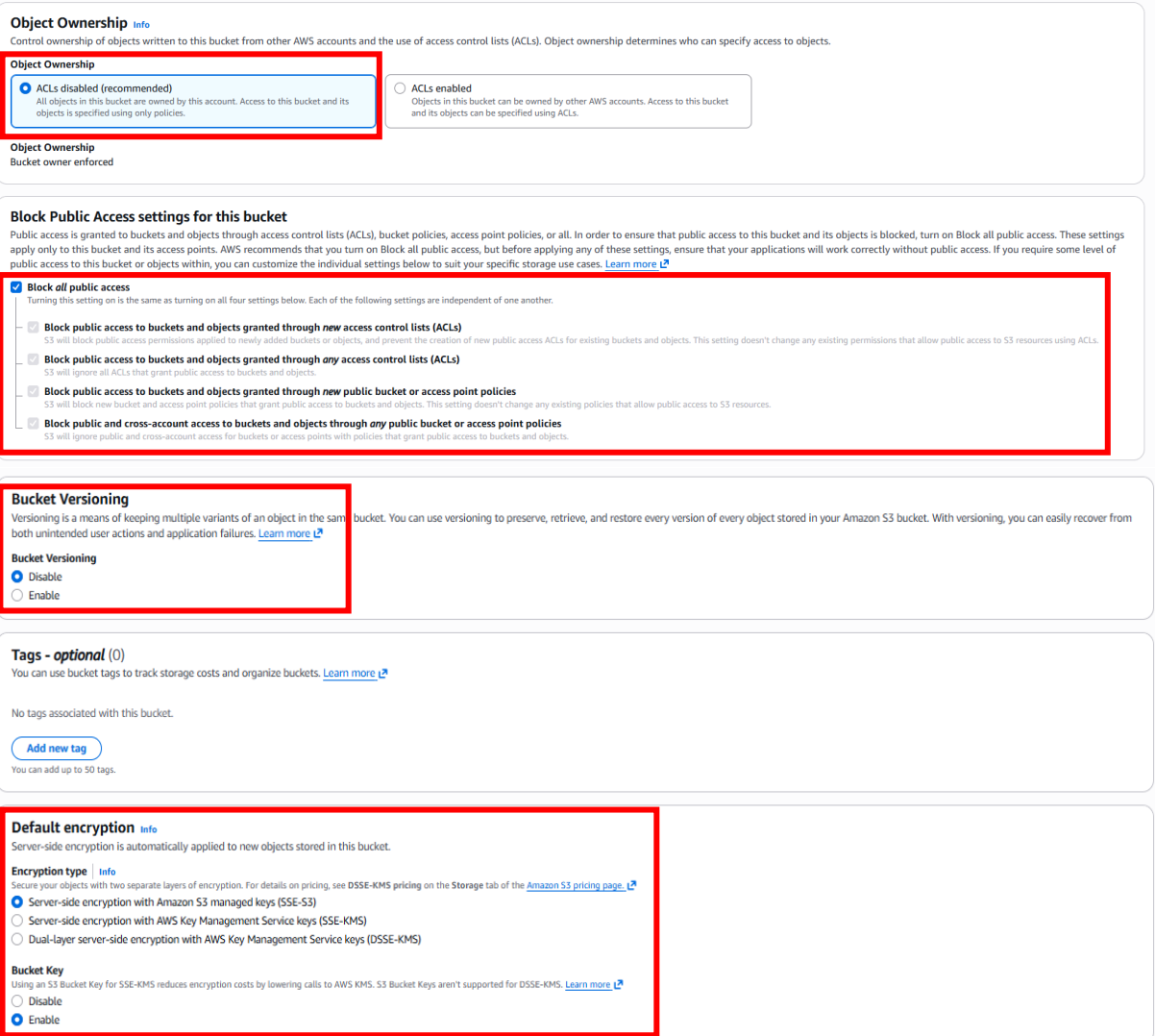
2 In the **AWS Management Console**, search for **“S3”** in the top search bar and click **Buckets**

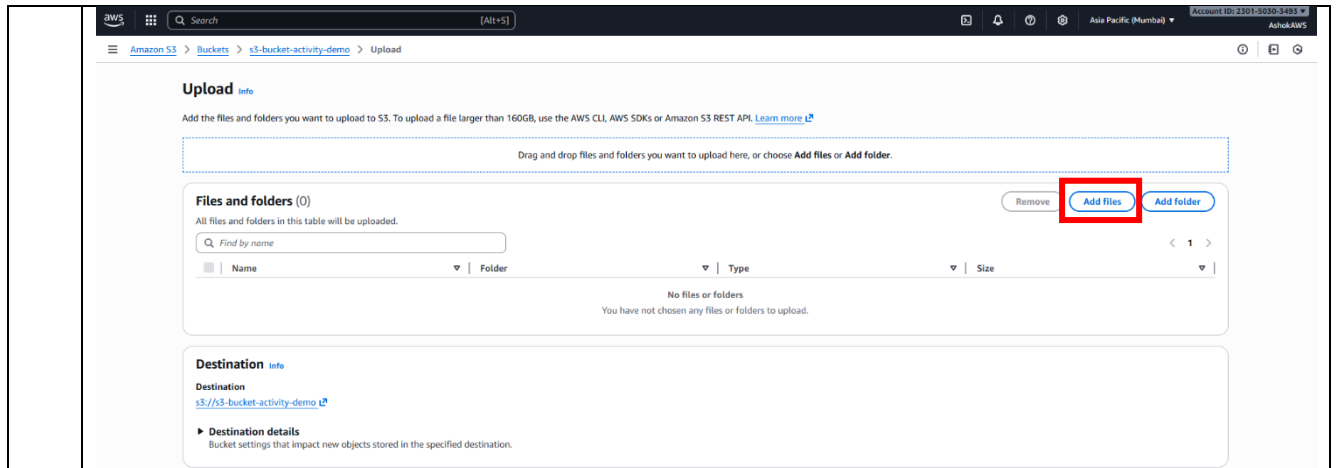


3 Click **Create Button** to create the bucket

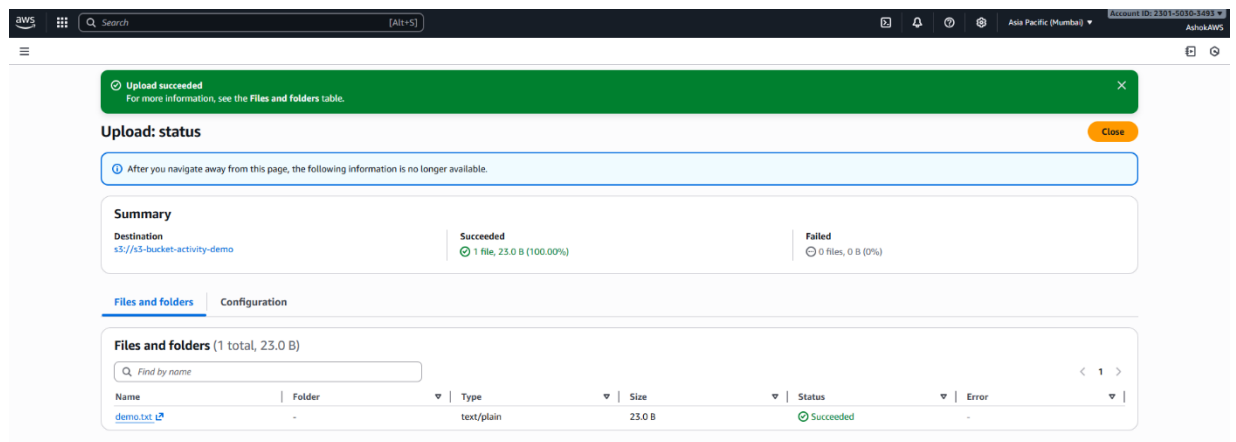


4 Enter a **unique bucket name** and choose the **AWS Region** closest to your users or resources.

	
5	Configure Bucket Settings for Object Ownership, Block Public Access settings, Bucket versioning and default encryption 
6	Review the settings and click Create bucket, success message will be displayed



8 A success message will be displayed after upload/add files.



Thus, the **S3 bucket is created and configured successfully**

Activity 2 - Design and implement a sample schema on Amazon RDS

2.1 Prerequisite

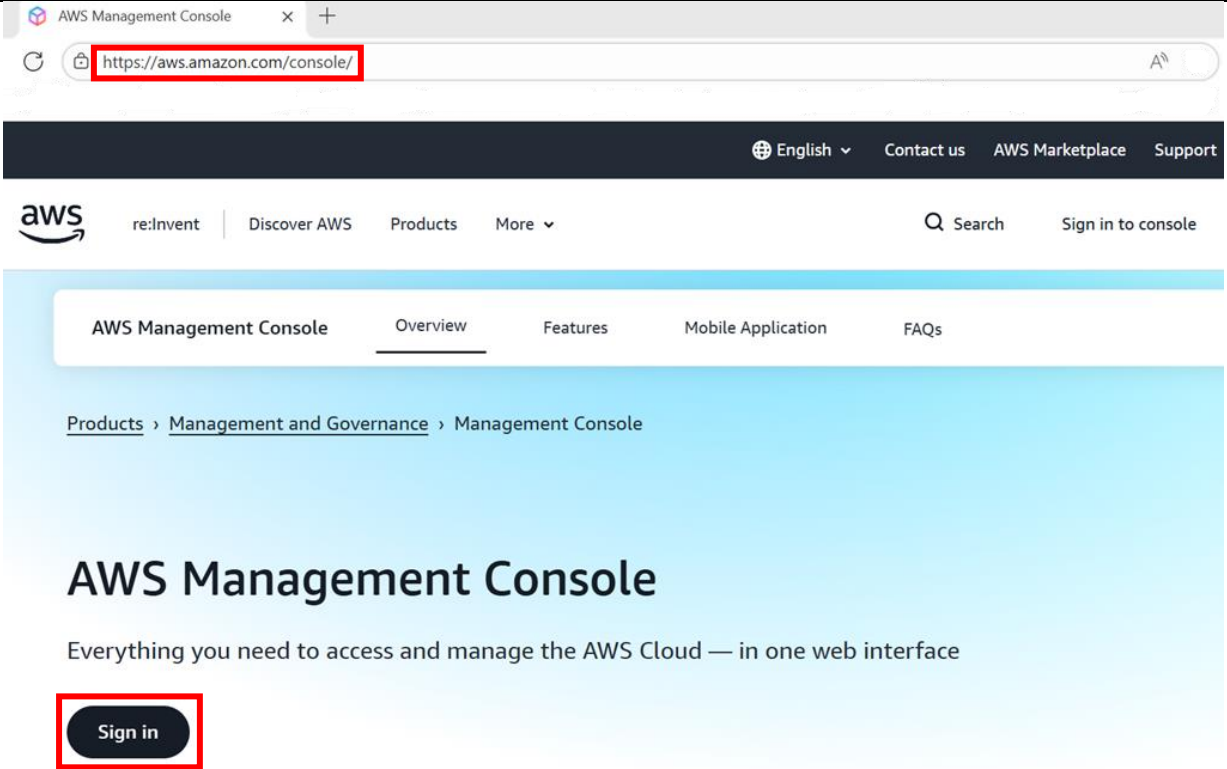
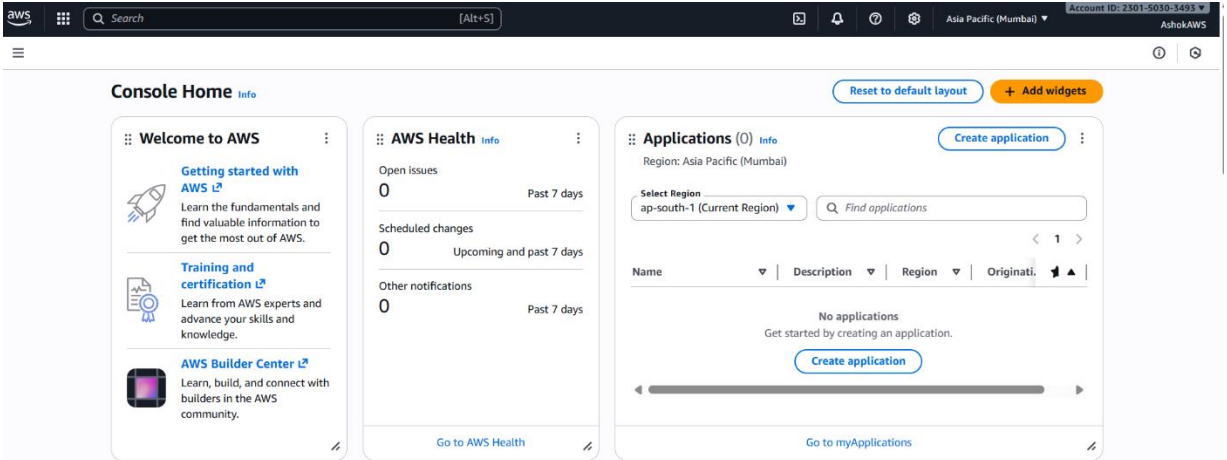
- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

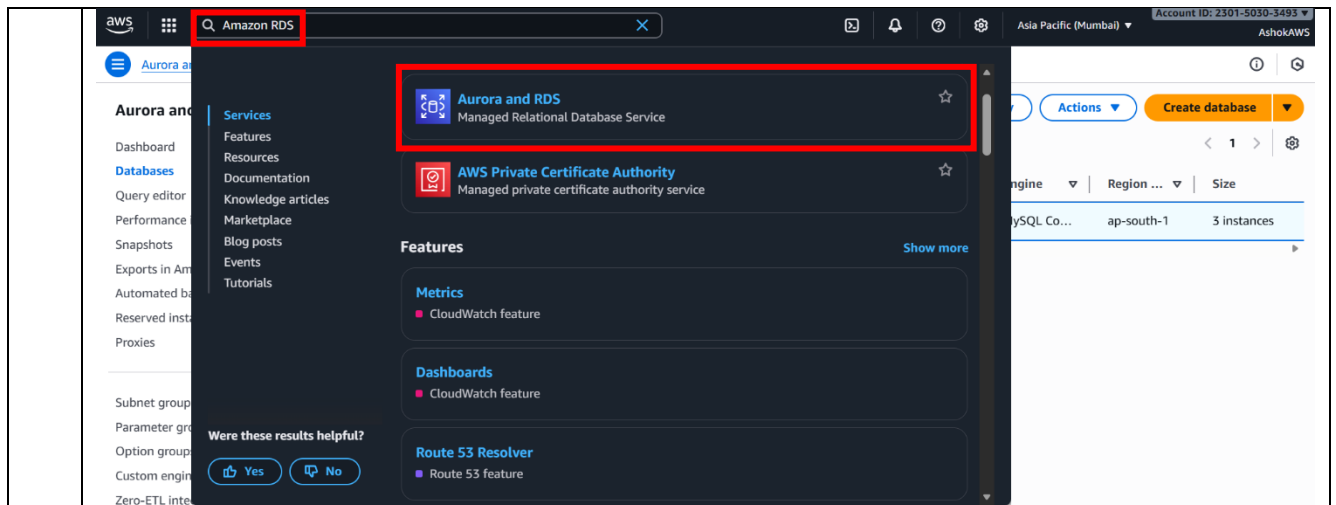
2.2 Scenario

Design and implement a sample schema on Amazon RDS

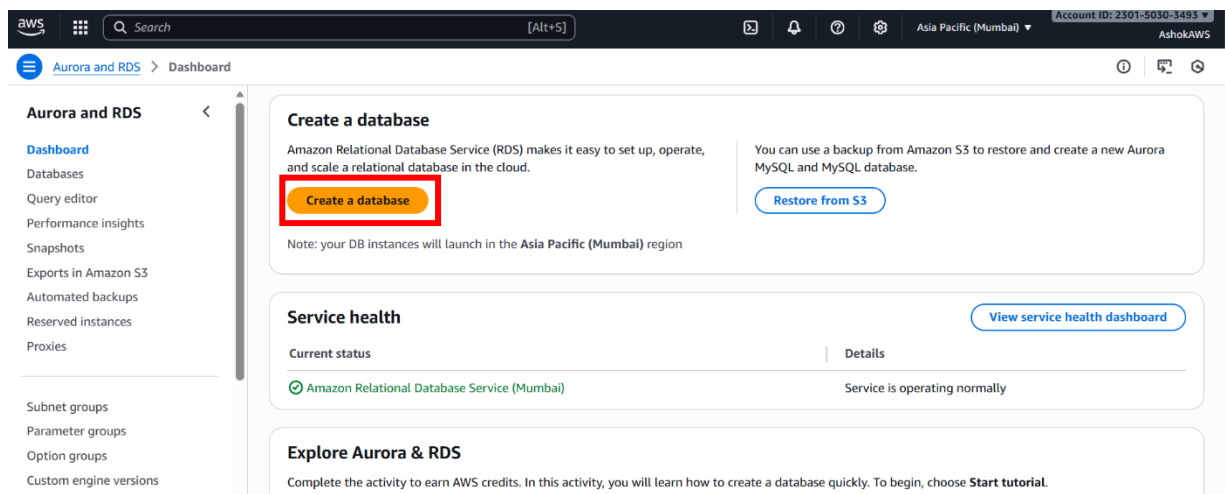
2.3 Steps

1	Log in to AWS Console • Visit https://aws.amazon.com/console • Sign in with your credentials
---	--

	 After Sign in, the following page will be displayed: 
2	In the AWS Management Console, search for “Amazon RDS” in the top search bar and click Aurora and RDS



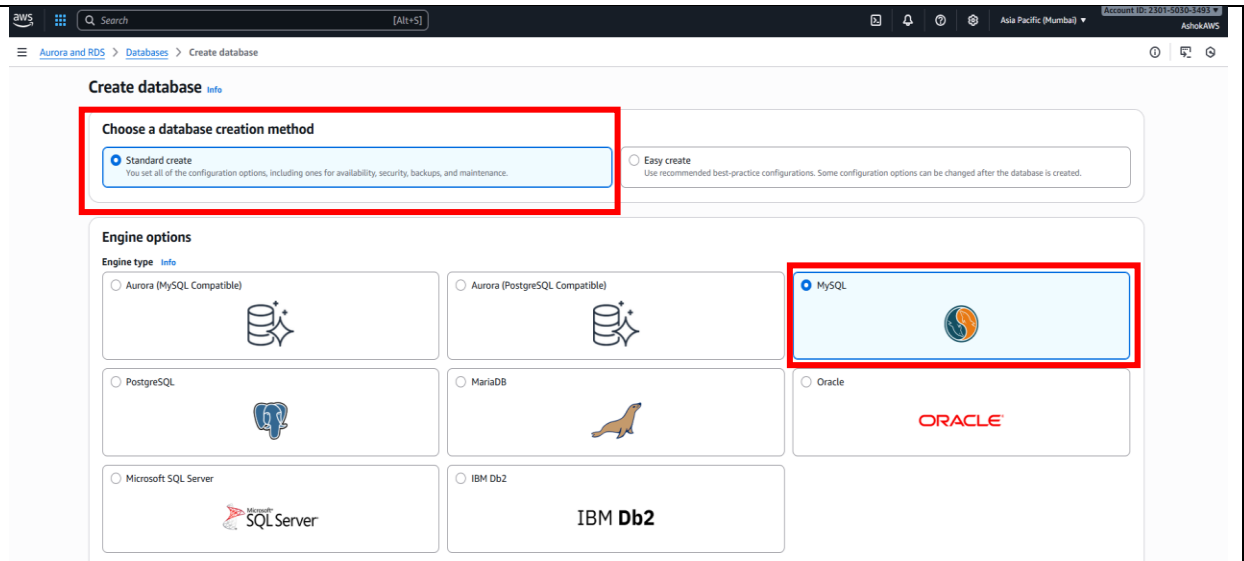
3 Click **“Create a database”** to create the database



4 Follow the steps below to create the database:

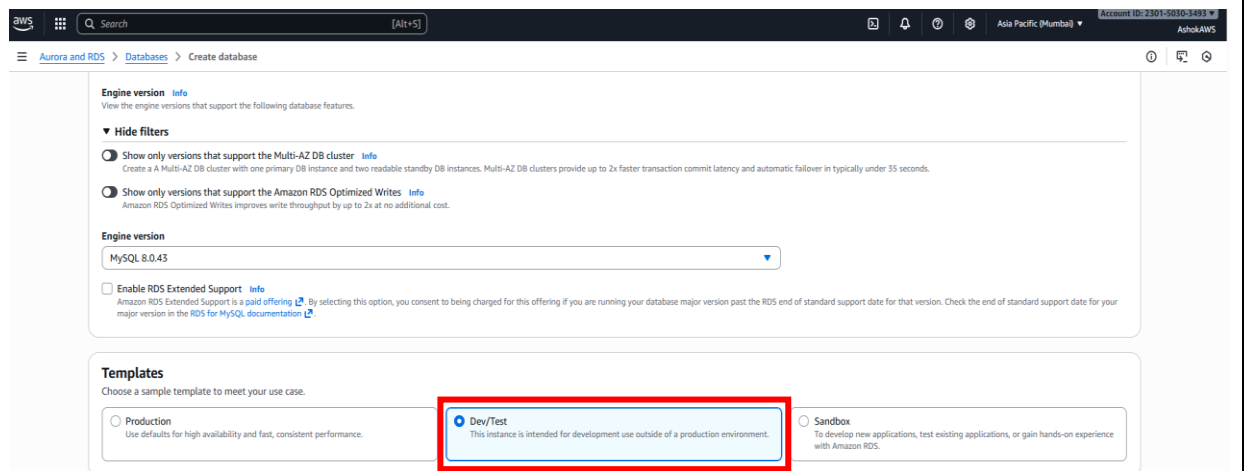
Step 4.1

- Under **Create database** → Click **“Standard create”** option from the database creation method
- Under **Engine options** → Click the **Engine type** as **“MySQL”**



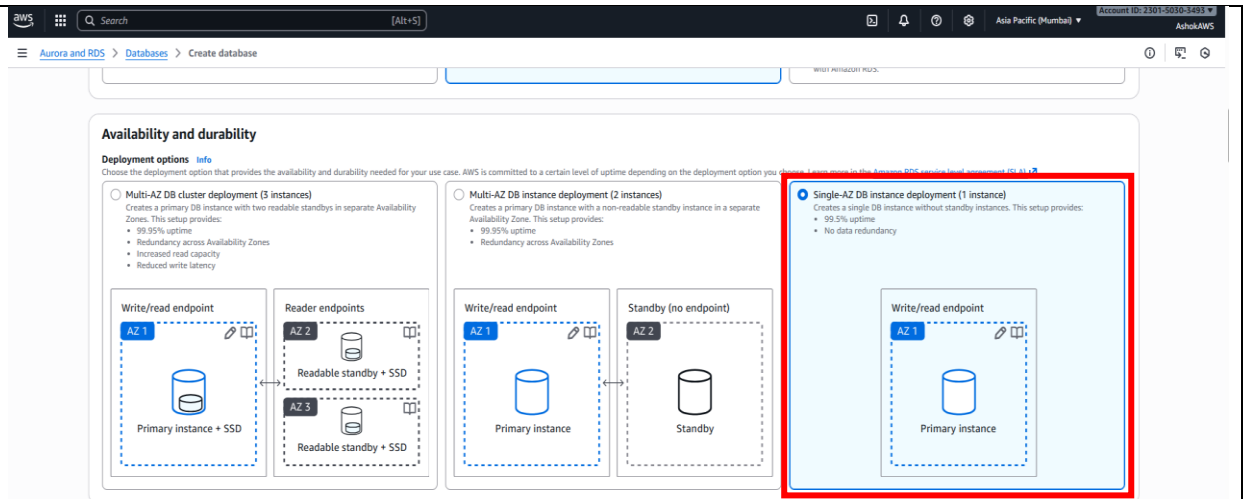
Step 4.2

Under **Templates** → Click “**Dev/Test**” option from the templates



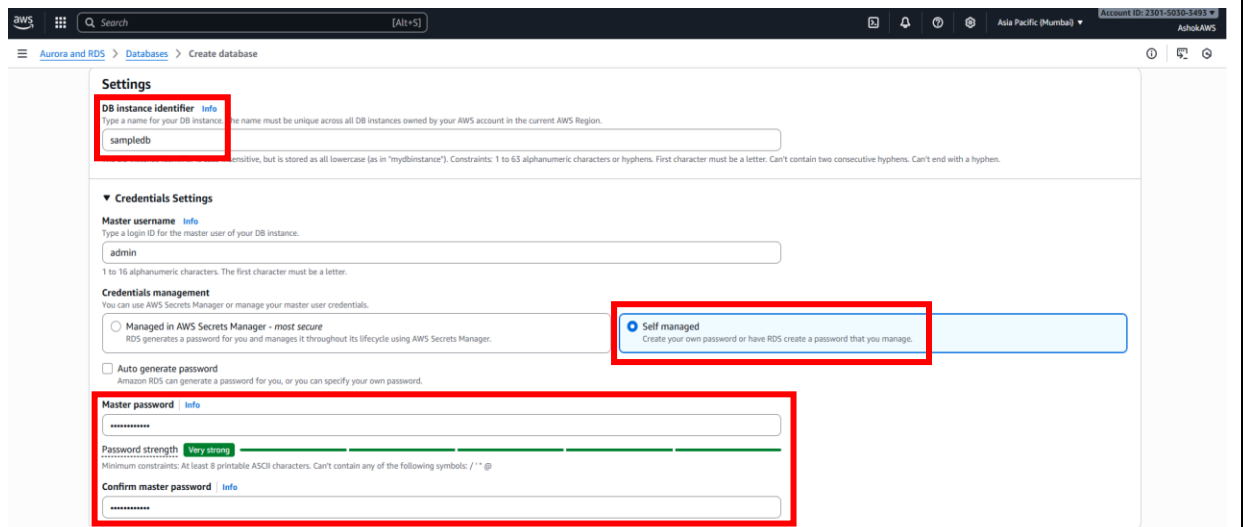
Step 4.3

Under **Availability and durability** (deployment options) → Click “**Single-AZ DB instance deployment (1 instance)**” for the activity



Step 4.4

- Under **Settings** → **DB instance identifier** → give a name for the DB instance (e.g., **sampledb**)
- Under **Credentials Settings** → **Credential management** → Click “**Self managed**” → **Master Password** → give a strong Password and confirm



Step 4.5

Under **Instance Configuration** → Click “**Burstable classes (includes t classes)**” → select **db.t3.micro** from the drop-down list

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ **Hide filters**

☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ **Burstable classes (includes t classes)**

db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Step 4.6

Under **Storage** → Choose the **Storage type** as **General Purpose SSD (gp3)** and **Allocated storage** as **20**

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)
Performance scales independently from storage

Allocated storage [Info](#)
20
Minimum: 20 GiB, Maximum: 6,144 GiB

Provisioned IOPS [Info](#)
5000
IOPS
Baseline IOPS of 3,000 IOPS is included for allocated storage less than 400 GiB.

Storage throughput [Info](#)
125
MiBps
Baseline storage throughput of 125 MiBps is included for allocated storage less than 400 GiB.

To provision additional IOPS and throughput, increase the allocated storage to 400 GiB or greater.

▼ **Additional storage configuration**

Storage autoscaling [Info](#)
Provides dynamic scaling support for your database's storage based on your application's needs.

☒ **Enable storage autoscaling**
Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Step 4.7

Under **Connectivity** → Click “**Don’t connect to an EC2 compute resource**” and Choose **VPC** as **Default VPC**

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-08b53f19aa4c1c1ab)
3 Subnets, 3 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

ⓘ After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-08b53f19aa4c1c1ab
3 Subnets, 3 Availability Zones

Public access [Info](#)

☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

Step 4.8

Under **VPC security group (firewall)** → Click **“Create new”** and give a name for the VPC security group name (e.g., *rds-sample-sg*)

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ **Choose existing**
Choose existing VPC security groups

☒ **Create new**
Create new VPC security group

New VPC security group name
rds-sample-sg

Availability Zone [Info](#)
No preference

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ **Create an RDS Proxy** [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061

If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

Database port [Info](#)
TCP/IP port that the database will use for application connections.

3306

Step 4.9

Under **Monitoring** → Click **“Database Insights – Standard”**

Monitoring [info](#)
Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. Database Insights pricing is separate from RDS monthly estimates. See [Amazon CloudWatch pricing](#).

☐ Database Insights - Advanced

- Retains 15 months of performance history
- Flux-level monitoring
- Integration with CloudWatch Application Signals

☒ **Database Insights - Standard**

▼ **Additional monitoring settings**
Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring
Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports
Select the log types to publish to Amazon CloudWatch Logs

☐ Audit log

☐ Error log

☐ General log

☐ iam-db-auth-error log

☐ Slow query log

IAM role
The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

Step 4.10

- Under **Additional configuration** → **Database options** → give a name for the initial database name (e.g., *school*)
- Under **Backup** → “Check the option: **Enable automated backup**” and give number of days under **Backup retention period** (e.g., 5)

▼ **Additional configuration**
Database options, encryption turned on, backup turned on, backtracks turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [info](#)
school

Amazon RDS does not create a database.

DB parameter group [info](#)
default.mysql8.0

Option group [info](#)
default-mysql-8-0

Backup

☒ **Enable automated backup**
Creates a point-in-time snapshot of your database.

Please note that automated backups are currently supported for InnoDB storage engine only. If you are using MySQL, refer to details [here](#).

Backup retention period [info](#)
The number of days (1-35) for which automatic backups are kept.
5 days

Backup window [info](#)
The daily time range (in UTC) during which RDS takes automated backups.

☐ Choose a window

☒ No preference

Step 4.11

After the above-mentioned steps, click “**Create database**” button

Database group: defaultgroup-01

Backup

Enable automated backup

Please note that automated backups are currently supported for MySQL engine only. If you are using MySQL, refer to [MySQL](#).

Backup retention period: 1

Backup window: No performance

Backup encryption

Enable encryption

DB instance class: db.t3.micro

Estimated monthly costs

Resource	Unit	Price
DB instance	1 instance	\$45.07 USD
Storage	100 GB	\$0.10 USD
Backup storage	100 GB	\$0.10 USD
Backup storage	100 GB	\$0.10 USD

Create database

AWS will now begin creating the instance which might take a few minutes to launch.

Creating database sampledb

Your database might take a few minutes to launch. You can use settings from sampledb to simplify configuration of suggested database add-ons while we finish creating your DB for you.

DB identifier	Status	Role	Engine	Region	Size	Recommendations	CPU	Current
sampledb	Creating	Instance	MySQL Co...	ap-south-1a	db.t3.micro			

A success message will be displayed after the database is created.

Successfully created database sampledb

You can use settings from sampledb to simplify configuration of suggested database add-ons while we finish creating your DB for you.

DB identifier	Status	Role	Engine	Region	Size	Recommendations	CPU	Current
sampledb	Available	Instance	MySQL Co...	ap-south-1a	db.t3.micro			

Once the DB status shows **“Available”**:

DB identifier	Status	Role	Engine	Region	Size	Recommendations	CPU	Current
sampledb	Available	Instance	MySQL Co...	ap-south-1a	db.t3.micro			

Follow the steps:

Step 4.12

Go to the **Connectivity & security** tab and find the **VPC security groups** → Click the link (shown).

The screenshot shows the AWS RDS console for an instance named 'sampledb'. The 'Connectivity & security' tab is selected. Under the 'Security' section, the 'VPC security groups' are listed as 'rds-sample-sg (sg-0cb7e57b300d70708)' with an 'Active' status. The 'Endpoint' section shows the instance is accessible on port 3306. The 'Networking' section shows the instance is in the 'ap-south-1a' availability zone, using VPC 'vpc-08b53f19aa4c1c1ab' and subnets 'subnet-088752c4cd12b299' and 'subnet-0e255ec78f52b0079'.

Step 4.13

In the **Security Groups** page → Click the ID under **Security group ID**

The screenshot shows the AWS EC2 console 'Security Groups' page. A search filter is applied to find security groups by attribute or tag. The 'Security group ID' column is highlighted with a red box, showing the ID 'sg-0cb7e57b300d70708'. The table lists the security group name 'rds-sample-sg', VPC ID 'vpc-08b53f19aa4c1c1ab', and description 'Created by RDS management console'.

Step 4.14

Under **Inbound rules** tab → Click **“Edit inbound rules”** → Click **“Add rule”** and add the given details:

- **Type:** MySQL/Aurora (or PostgreSQL)
- **Port:** 3306 (for MySQL) or 5432 (for PostgreSQL)
- **Source:** My IP
- Click **“Save rules”**

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

sg-0cb7e57b300d70708 - rds-sample-sg

Details

Security group name
rds-sample-sg

Security group ID
sg-0cb7e57b300d70708

Description
Created by RDS management console

VPC ID
vpc-08b53f19aad41c1ab

Owner
230150303493

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0c2d8e11a7a2a1988	IPv4	MySQL/Aurora	TCP	3306	223.237.181.117/32	-

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID
sgr-0c2d8e11a7a2a1988

Type
MySQL/Aurora

Protocol
TCP

Port range
3306

Source
My IP

Description - optional

Delete

Add rule

Cancel Preview changes Save rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID
sgr-0c2d8e11a7a2a1988

Type
MySQL/Aurora

Protocol
TCP

Port range
3306

Source
My IP

Description - optional

Delete

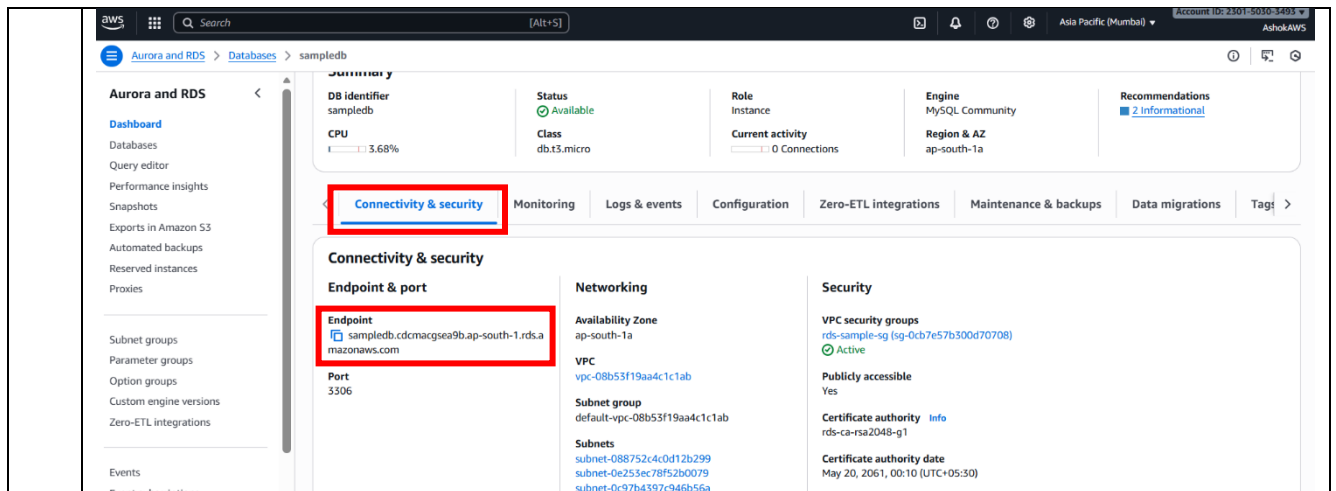
Add rule

Cancel Preview changes Save rules

This makes sure **only your computer** can connect to the database.

Step 4.15

After the above-mentioned steps, go back to your RDS instance → **Connectivity & security tab** → Copy the **Endpoint** (e.g., **sampleddb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com**)



You'll use the **Endpoint** to connect like this in **Command-Line Interface**:

```
mysql -h sampledb.cdcmagsea9b.ap-south-1.rds.amazonaws.com -u admin -p
```

Note: In **Step 4.10**, the database name **“school”** was created.

Let's create a simple **“school database”** with:

- **Students** (student_id, name, email)
- **Courses** (course_id, title, credits)
- **Enrollments** (enrollment_id, student_id, course_id, grade)

5 **Connect to the RDS Database**

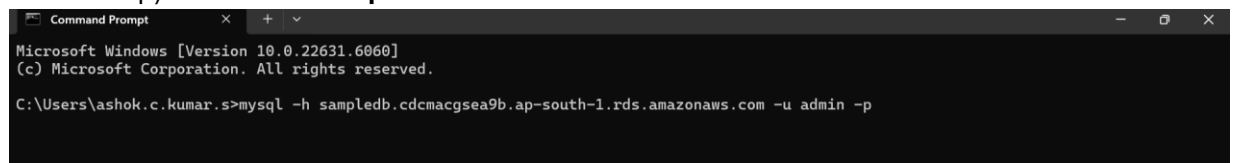
Once your RDS instance is available.

1. Find the **Endpoint** in the RDS console
(e.g., **sampledb.cdcmagsea9b.ap-south-1.rds.amazonaws.com**).
2. Use a SQL client such as:
 - **MySQL Workbench**
 - **pgAdmin**
 - **DBeaver**
 - Or the terminal (**mysql** or **psql** command-line client).

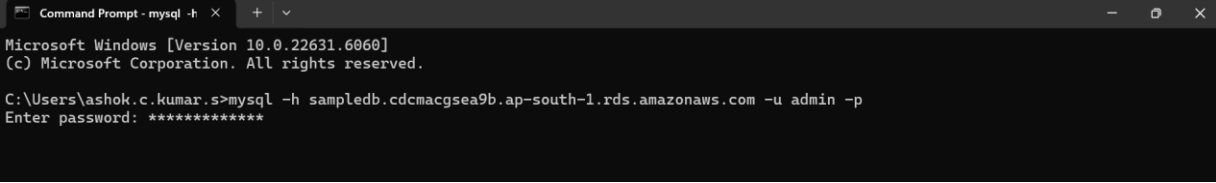
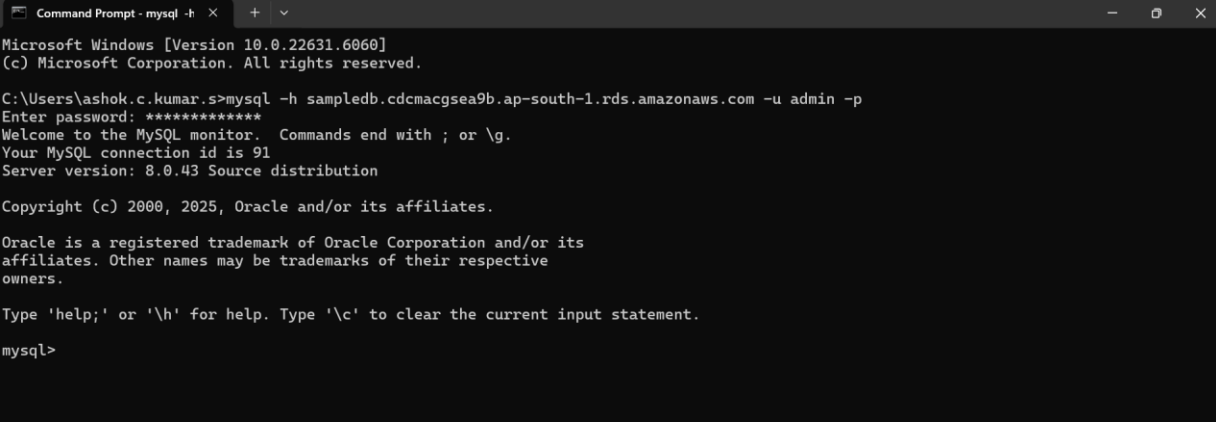
Here, we took **MySQL CLI** (Command-Line Interface)

Step 1: Search for **“cmd”** (**Command Prompt**) in Start menu

Step 2: Type the **Endpoint** (mysql -h sampledb.cdcmagsea9b.ap-south-1.rds.amazonaws.com -u admin -p) as shown in **Step 4.11**



Step 3: It will prompt for a **password**; enter the **Master password** given in **Step 4.4**

	 Step 4: After password is entered, click Enter key  Step 5: mysql command is ready for execution 
6	Create the Database and Schema Once connected, run SQL commands to create your schema.

```

mysql> CREATE DATABASE school;
Query OK, 1 row affected (0.07 sec)

mysql> USE school;
Database changed
mysql>
mysql> CREATE TABLE Students (
->     student_id INT AUTO_INCREMENT PRIMARY KEY,
->     name VARCHAR(100),
->     email VARCHAR(100) UNIQUE
-> );
Query OK, 0 rows affected (0.12 sec)

mysql>
mysql> CREATE TABLE Courses (
->     course_id INT AUTO_INCREMENT PRIMARY KEY,
->     title VARCHAR(100),
->     credits INT
-> );
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql> CREATE TABLE Enrollments (
->     enrollment_id INT AUTO_INCREMENT PRIMARY KEY,
->     student_id INT,
->     course_id INT,
->     grade CHAR(1),
->     FOREIGN KEY (student_id) REFERENCES Students(student_id),
->     FOREIGN KEY (course_id) REFERENCES Courses(course_id)
-> );
Query OK, 0 rows affected (0.12 sec)

```

Insert Sample Data

```

mysql> INSERT INTO Students (name, email)
-> VALUES ('Jack', 'jack@example.com'),
->         ('Eliza', 'eliza@example.com'),
->         ('William', 'william@example.com'),
->         ('Maria', 'maria@example.com');
Query OK, 4 rows affected (0.10 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Courses (title, credits)
-> VALUES ('Cloud Computing', 4),
->         ('Database Systems', 3),
->         ('Basic Programming', 4);
Query OK, 3 rows affected (0.08 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Enrollments (student_id, course_id, grade)
-> VALUES (1, 1, 'A'),
->         (2, 3, 'B'),
->         (3, 1, 'B');
Query OK, 3 rows affected (0.07 sec)
Records: 3  Duplicates: 0  Warnings: 0

```

Retrieve the inserted Sample Data:

```
mysql> SELECT * FROM Students;
+-----+-----+-----+
| student_id | name   | email                |
+-----+-----+-----+
| 1          | Jack   | jack@example.com     |
| 2          | Eliza  | eliza@example.com    |
| 3          | William| william@example.com  |
| 4          | Maria  | maria@example.com    |
+-----+-----+-----+
4 rows in set (0.06 sec)
```

```
mysql> SELECT * FROM Courses;
+-----+-----+-----+
| course_id | title           | credits |
+-----+-----+-----+
| 1          | Cloud Computing | 4       |
| 2          | Database Systems| 3       |
| 3          | Basic Programming| 4       |
+-----+-----+-----+
3 rows in set (0.06 sec)
```

```
mysql> SELECT * FROM Enrollments;
+-----+-----+-----+-----+
| enrollment_id | student_id | course_id | grade |
+-----+-----+-----+-----+
| 1             | 1          | 1         | A     |
| 2             | 2          | 3         | B     |
| 3             | 3          | 1         | B     |
+-----+-----+-----+-----+
3 rows in set (0.07 sec)
```

Test Your Schema

```
mysql> SELECT s.name, c.title, e.grade
-> FROM Enrollments e
-> JOIN Students s ON e.student_id = s.student_id
-> JOIN Courses c ON e.course_id = c.course_id;
+-----+-----+-----+
| name   | title           | grade |
+-----+-----+-----+
| Jack   | Cloud Computing | A     |
| Eliza  | Basic Programming| B     |
| William| Cloud Computing | B     |
+-----+-----+-----+
3 rows in set (0.09 sec)
```

Thus, the **sample schema** is designed and implemented on Amazon RDS successfully

Activity 3 – Create schema on DynamoDB

3.1 Prerequisite

- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

3.2 Scenario

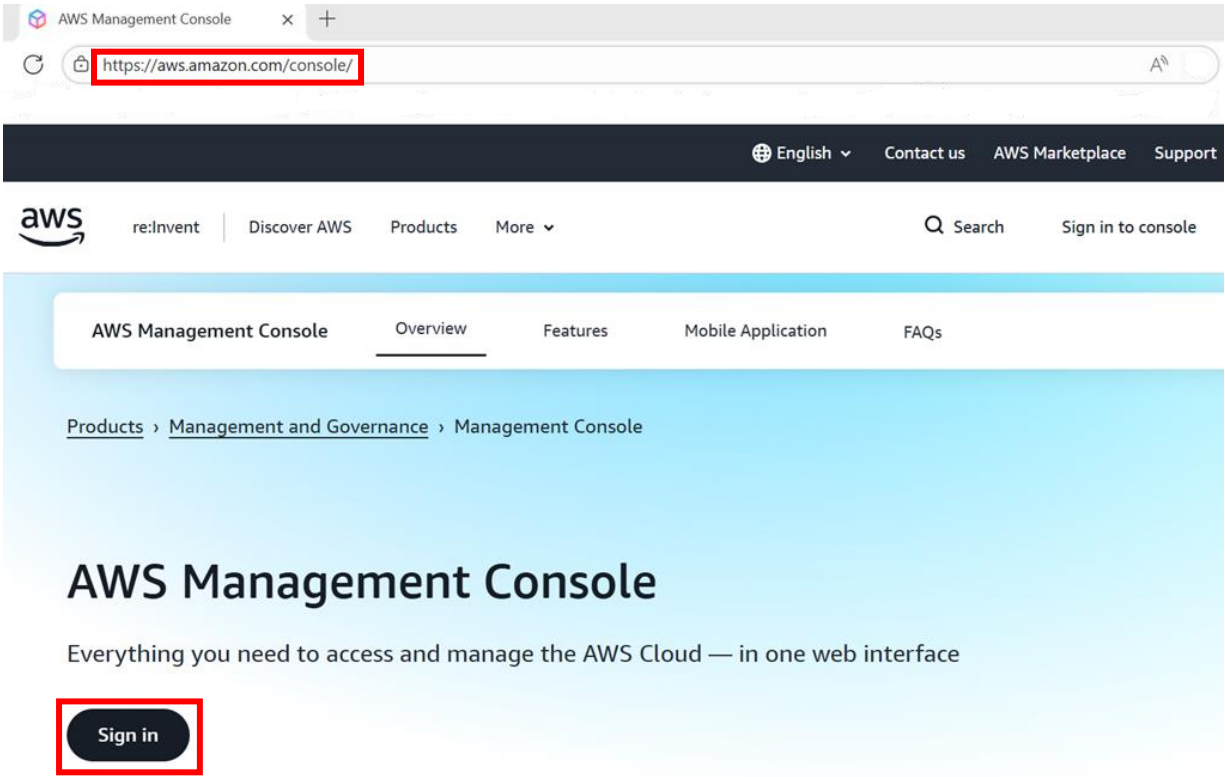
Create schema on DynamoDB, storing and retrieving data

3.3 Steps

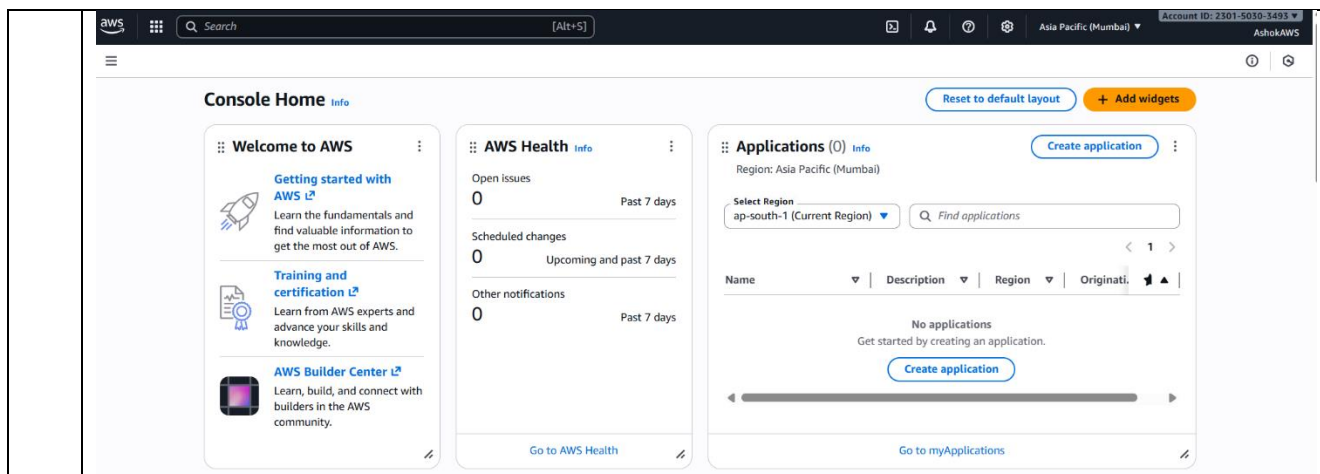
1

Log in to AWS Console

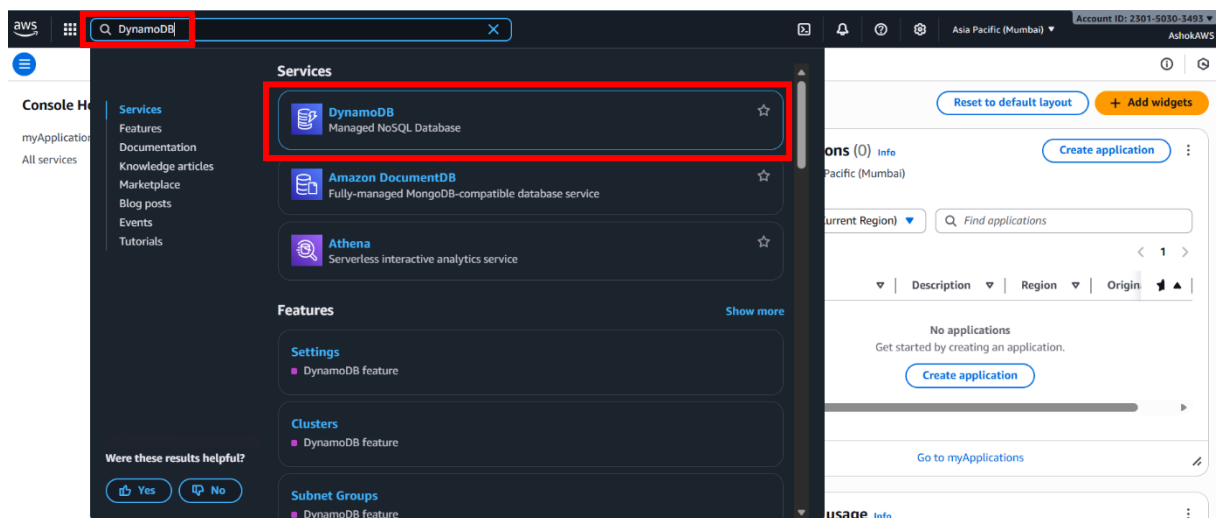
- Visit <https://aws.amazon.com/console>
- Sign in with your credentials



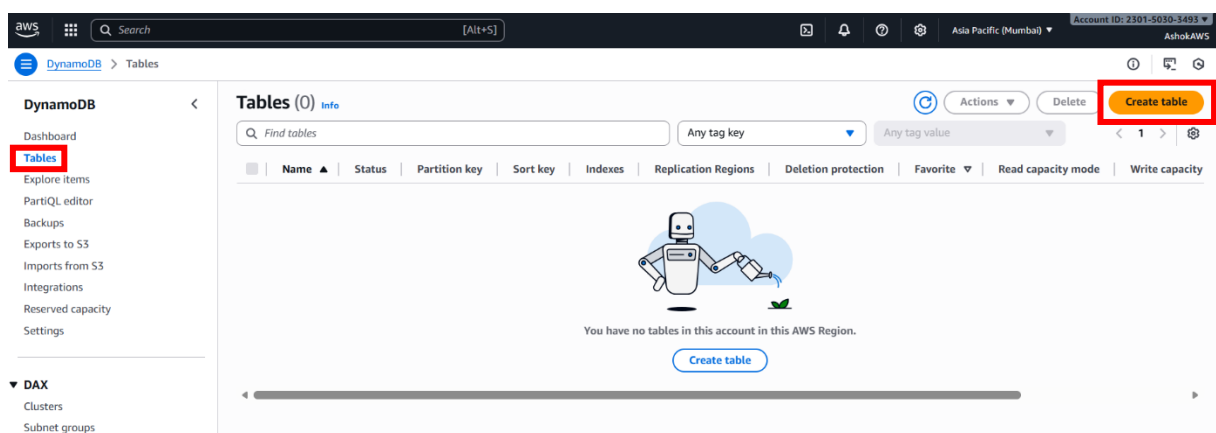
After **Sign in**, the following page will be displayed:



- 2 In the **AWS Management Console**, search for **“DynamoDB”** in the top search bar and click **DynamoDB**



- 3 In the left-navigation pane, click **Tables** and click the **“Create table”** button

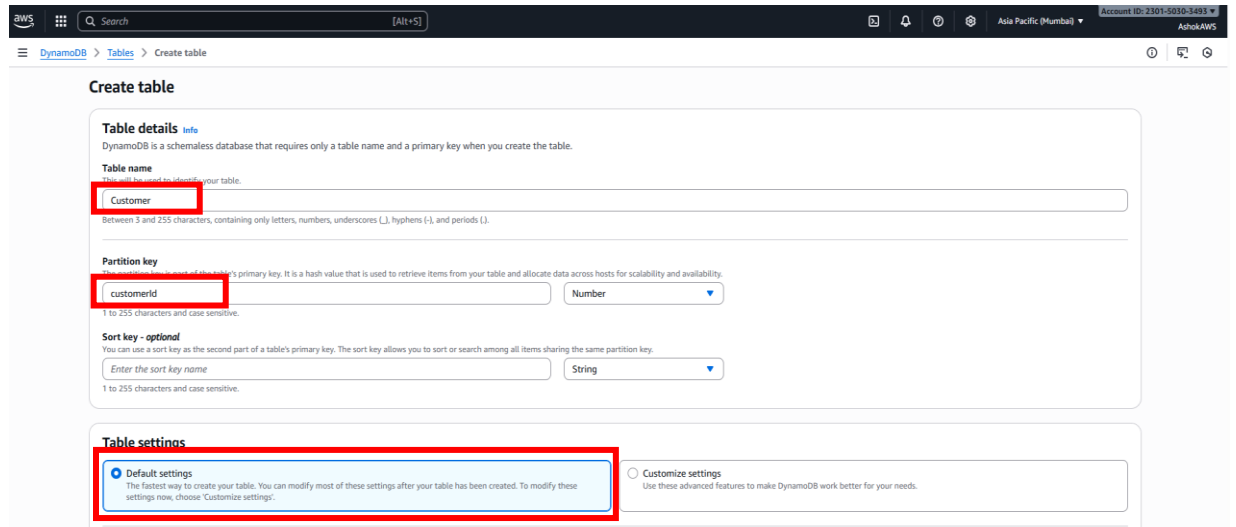


4

On clicking the **“Create table”** button, Enter **Table details – Table name** and **Partition key** and choose **Default settings** under Table settings

Step 4.1

- **Table name:** Enter a meaningful name
- **Partition key:** A mandatory attribute that uniquely identifies items
- Leave defaults and click **Create table**



Create table

Table details [Info](#)

DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

Table name

This will be used to identify your table.
Between 3 and 255 characters, containing only letters, numbers, underscores (_), hyphens (-), and periods (.).

Partition key

The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.
1 to 255 characters and case sensitive.

Sort key - optional

You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.
Enter the sort key name

1 to 255 characters and case sensitive.

Table settings

☒ **Default settings**
The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose "Customize settings".

☐ **Customize settings**
Use these advanced features to make DynamoDB work better for your needs.

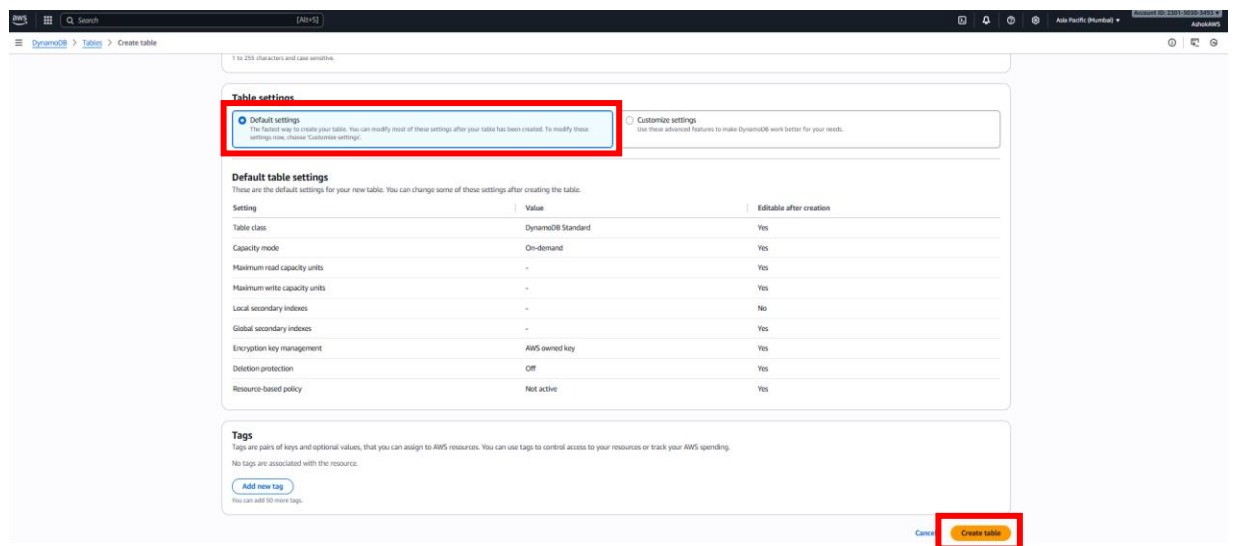


Table settings

☒ **Default settings**
The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose "Customize settings".

☐ **Customize settings**
Use these advanced features to make DynamoDB work better for your needs.

Default table settings

These are the default settings for your new table. You can change some of these settings after creating the table.

Setting	Value	Editable after creation
Table class	DynamoDB Standard	Yes
Capacity mode	On-demand	Yes
Maximum read capacity units	-	Yes
Maximum write capacity units	-	Yes
Local secondary indexes	-	No
Global secondary indexes	-	Yes
Encryption key management	AWS owned key	Yes
Deletion protection	Off	Yes
Resource-based policy	Not active	Yes

Tags

Tags are pairs of keys and optional values, that you can assign to AWS resources. You can use tags to control access to your resources or track your AWS spending.
No tags are associated with the resource.

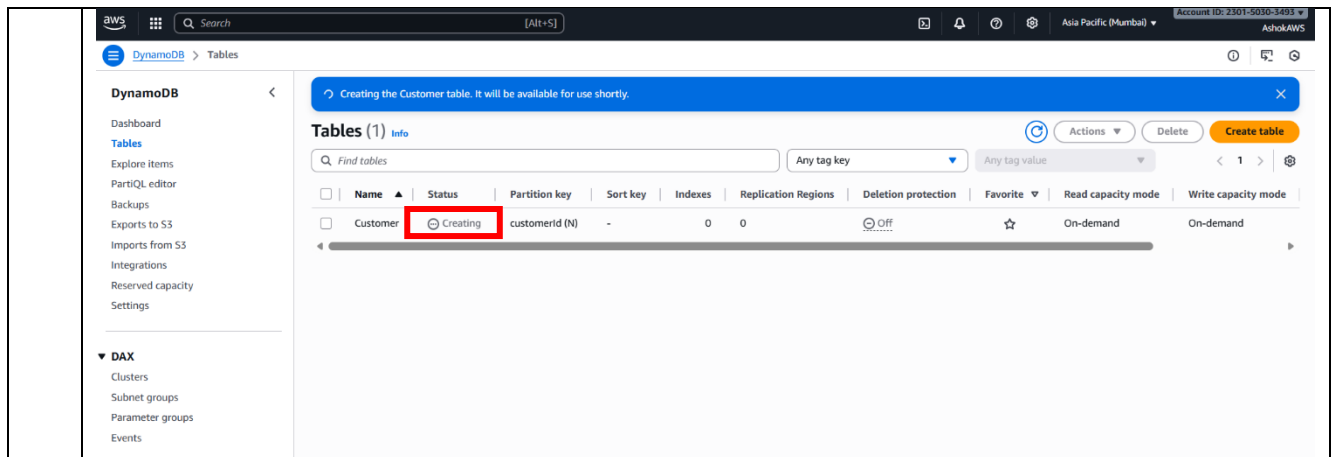
[Add new tag](#)

You can add 50 more tags.

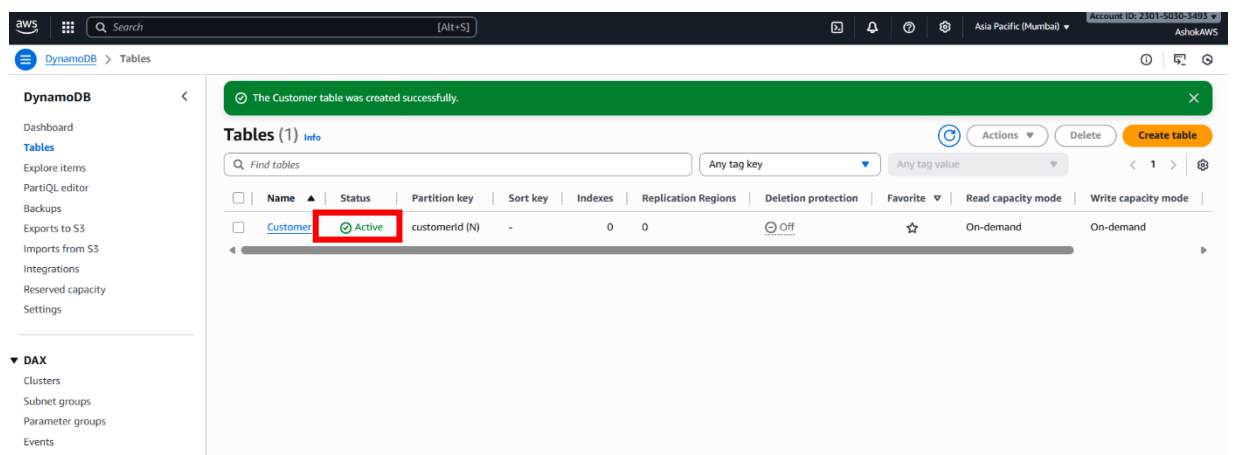
[Cancel](#) [Create table](#)

Step 4.2

- After clicking **Create table** button, the table will show up in the **“Tables”** list
- Initially the status will be **“Creating”**

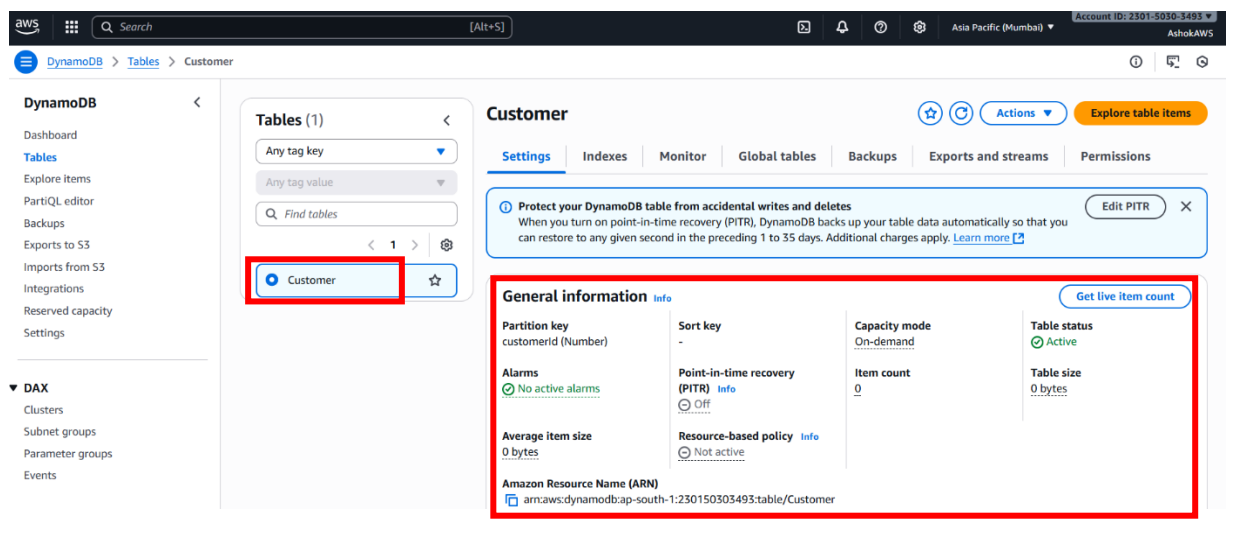


Later, it becomes **“Active”** once ready.



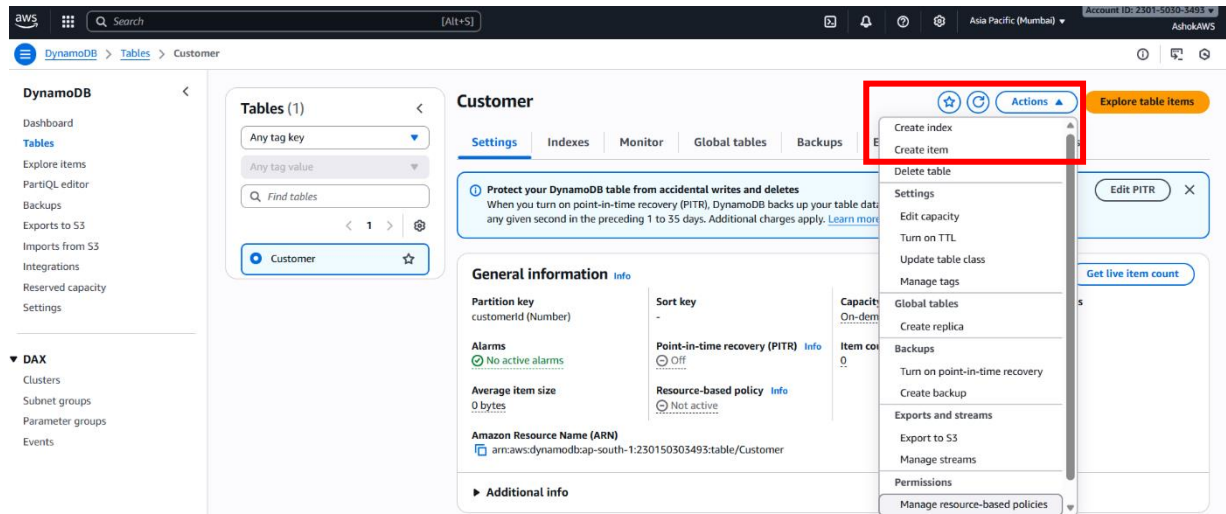
Step 4.3

Once the status is **Active**, click on the table name to view the below details:

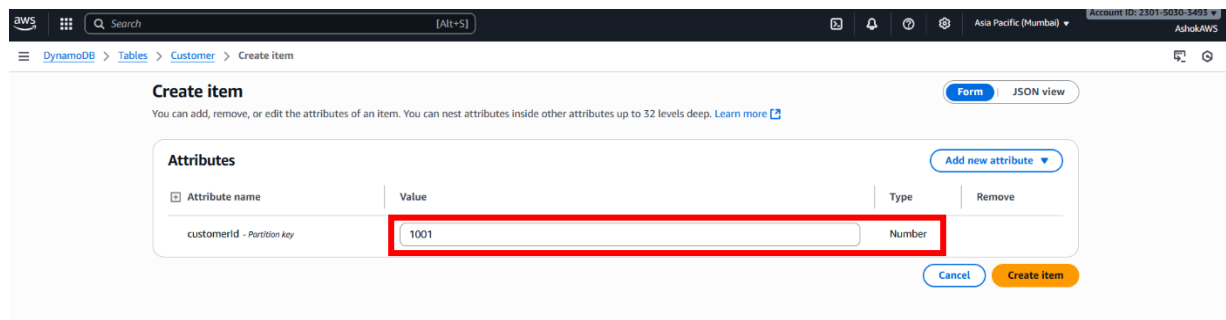


Step 4.4 Store items

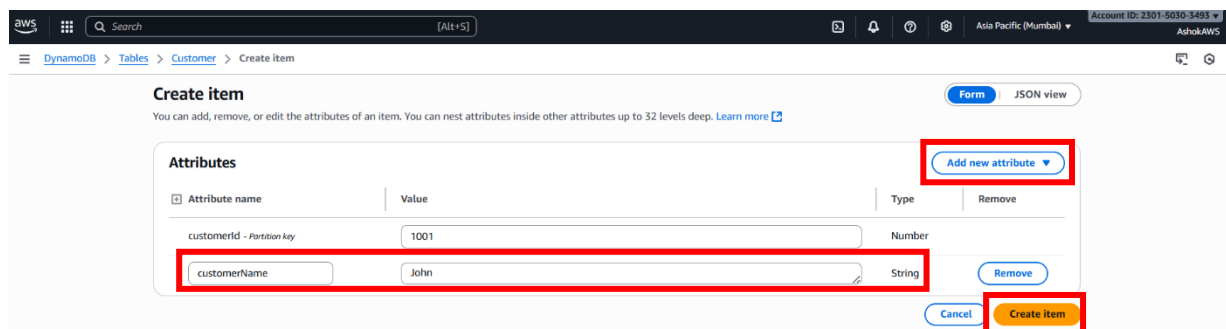
Click **Actions** → **Create item**



Under **Create item**, add the Value (based on the Type **e.g., Number, String, etc.,**) for the given Attribute name



If required, add a new attribute by specifying its Name and Value, and select the Type from the “Add new attribute” drop-down list.



Step 4.5 Retrieve items

- After adding the details, click the **“Create Item”** button
- A message will indicate whether the item was saved successfully
- The items will be displayed in the console

The screenshot shows the AWS DynamoDB console interface. A green banner at the top indicates 'The item has been saved successfully'. The left sidebar shows the 'DynamoDB' menu with options like 'Dashboard', 'Tables', 'Explore items', 'PartiQL editor', 'Backups', 'Exports to S3', 'Imports from S3', 'Integrations', 'Reserved capacity', and 'Settings'. The main area is titled 'Customer' and shows a 'Scan or query items' section. Below this, a 'Completed' message states 'Items returned: 1, Items scanned: 1, Efficiency: 100%, RCUs consumed: 2'. A table titled 'Table: Customer - Items returned (1)' is displayed, showing one item with the primary key 'customerid (Number)' set to '1001' and the attribute 'customerName' set to 'John'. The table is highlighted with a red border.

customerid (Number)	customerName
1001	John

Thus, the **schema is created on DynamoDB**, with **data stored and retrieved successfully**.

Activity 4 – Create a VPC with public and private subnets

4.1 Prerequisite

- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

4.2 Scenario

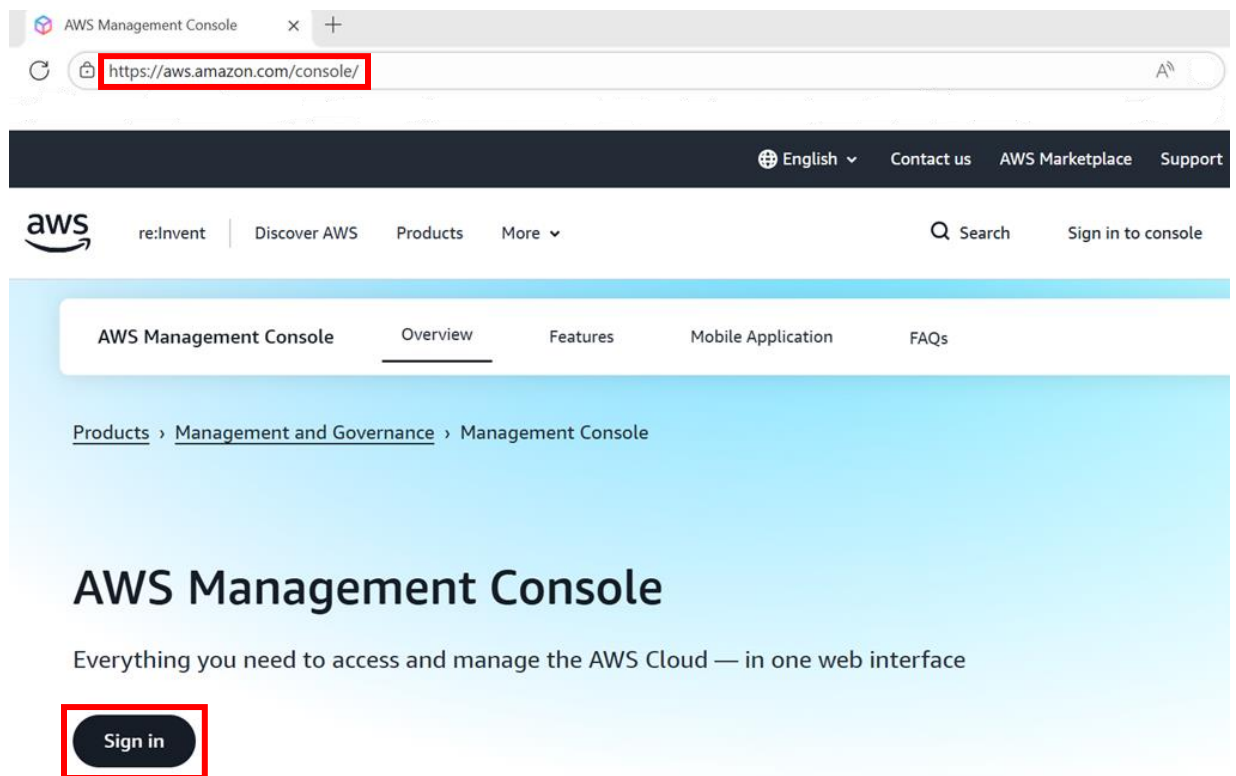
Create a VPC with public and private subnets

4.3 Steps

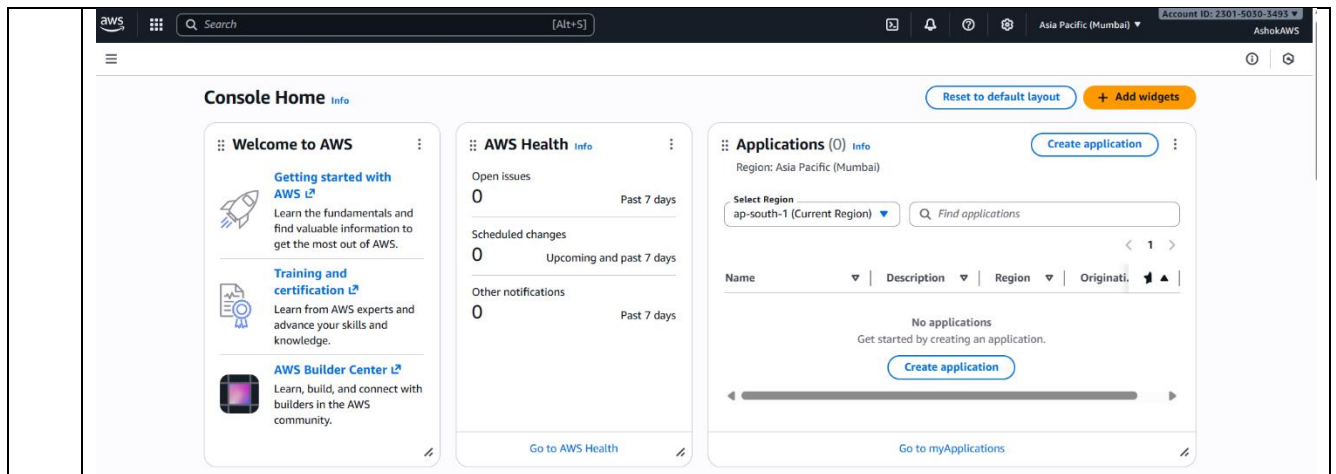
1

Log in to AWS Console

- Visit <https://aws.amazon.com/console>
- Sign in with your credentials



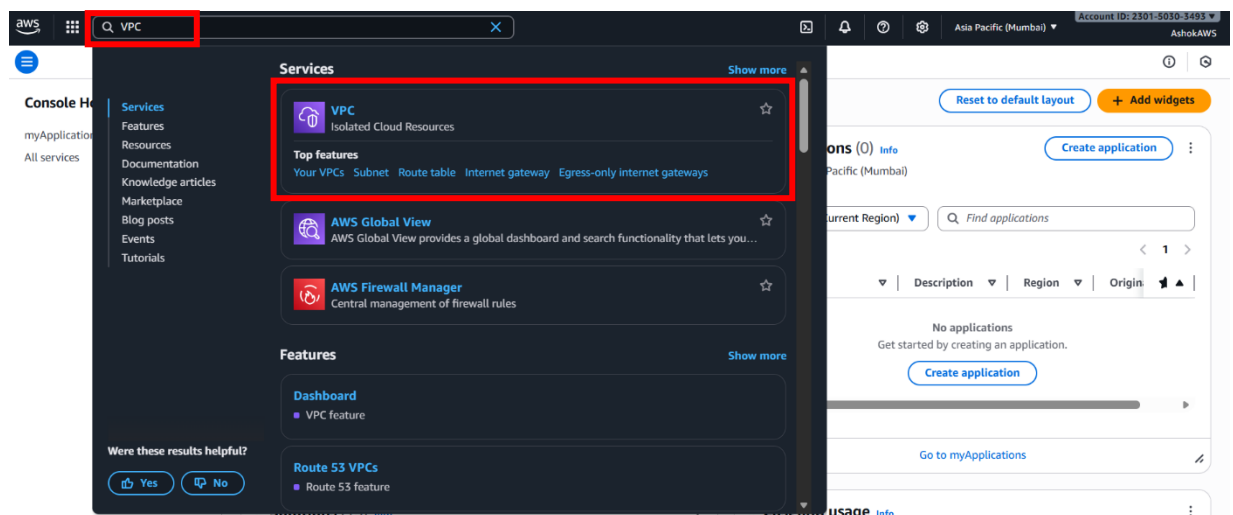
After **Sign in**, the following page will be displayed:



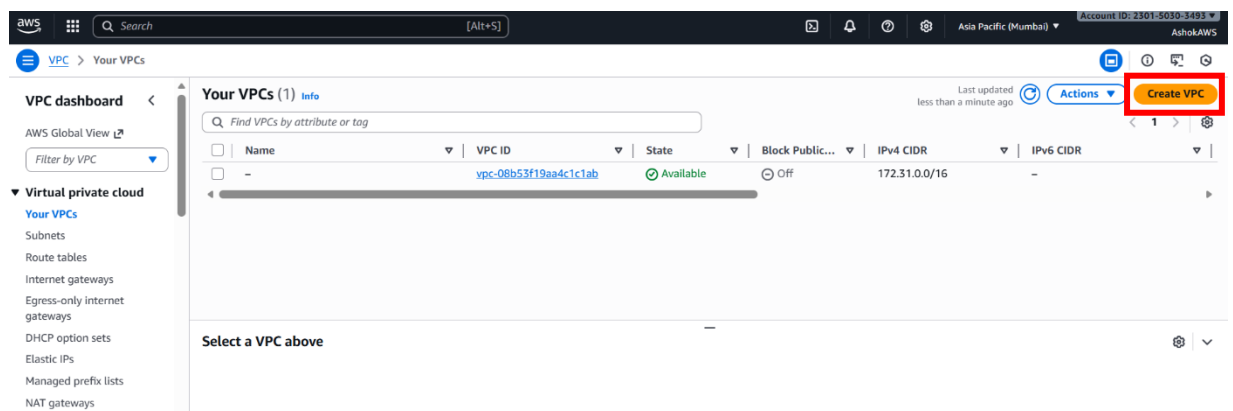
2

Create the VPC

Step 2.1 In the AWS Management Console, search for “VPC” in the top search bar and click “Your VPCs”



Step 2.2 Click “Create VPC” button



Step 2.3 Add the details shown below for the Name tag and IPv4 CIDR. Leave all other settings as default, then click the **“Create VPC”** button

The screenshot shows the 'Create VPC' page in the AWS Management Console. The 'Name tag - optional' field is highlighted with a red box and contains the text 'MyVPC'. The 'IPv4 CIDR' field is also highlighted with a red box and contains the text '10.0.0.0/16'. The 'Create VPC' button at the bottom right is highlighted with a red box.

A success message will be displayed after creating the VPC.

The screenshot shows the 'VPC dashboard' page in the AWS Management Console. A green success message at the top reads 'You successfully created vpc-078ce8a2d415442a7 / MyVPC'. The page shows details for the VPC, including its ID, state (Available), and various settings like DNS resolution, Main network ACL, and IPv6 CIDR. The 'Resource map' section at the bottom shows the VPC and its associated subnets, route tables, and network connections.

3 Create Subnets - Create at least one public and one private subnet.

Step 3.1 Create Public Subnet

Click **VPC → Subnets → Create Subnet**

VPC dashboard <

AWS Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections

Subnets (3) Info

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
-	subnet-0c97b4397c946b56a	Available	vpc-08b53f19aa4c1c1ab	Off	172.31.0.0/20	-
-	subnet-0e253ec78f52b0079	Available	vpc-08b53f19aa4c1c1ab	Off	172.31.16.0/20	-
-	subnet-088752c4cd12b299	Available	vpc-08b53f19aa4c1c1ab	Off	172.31.32.0/20	-

Last updated 7 minutes ago

Actions Create subnet

Select a subnet

Add the details shown below, then click “Create subnet” button

Create subnet Info

VPC

VPC ID

Choose a VPC to create the subnet in.

vpc-07b08a2d415442a7 | MyVPC

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of Name and a value that you specify.

PublicSubnet

The name can be up to 256 characters long.

Availability Zone

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

Tags - optional

Remove

Add new subnet

Create subnet

A success message will be displayed after creating subnet (PublicSubnet).

VPC dashboard <

AWS Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Subnets (1) Info

Find subnets by attribute or tag

Subnet ID : subnet-00bea82fad828183c

Clear filters

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
PublicSubnet	subnet-00bea82fad828183c	Available	vpc-07b08a2d415442a7 MyV...	Off	10.0.1.0/24	-

Last updated less than a minute ago

Actions Create subnet

Select a subnet

Step 3.2 Create Private Subnet

Click **Subnets** → **Create Subnet**

The screenshot shows the AWS VPC console with the 'Subnets' page selected. On the left sidebar, 'Subnets' is highlighted under 'Virtual private cloud'. The main area displays a table of existing subnets. In the top right corner, the 'Create subnet' button is highlighted with a red box.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 C...
-	subnet-0c97b4397c946b56a	Available	vpc-08b53f19aa4c1c1ab	Off	172.31.0.0/20	-	-
-	subnet-0e253ec78f52b0079	Available	vpc-08b53f19aa4c1c1ab	Off	172.31.16.0/20	-	-
-	subnet-08875264cd12b299	Available	vpc-08b53f19aa4c1c1ab	Off	172.31.32.0/20	-	-
PublicSubnet	subnet-00bea82fad828183c	Available	vpc-078ce8a2d415442a7 MyV...	Off	10.0.1.0/24	-	-

Add the details shown below, then click “Create subnet” button

The screenshot shows the 'Create subnet' form in the AWS VPC console. The form fields are highlighted with a red box. The 'Create subnet' button is highlighted in the bottom right corner.

Create subnet

VPC
 VPC ID: vpc-078ce8a2d415442a7 (MyVPC)

Associated VPC CIDRs
 IPv4 CIDRs: 10.0.0.0/16

Subnet settings
 Specify the CIDR block and Availability Zone for the subnet.

Subnet 1 of 1
 Subnet name: PrivateSubnet
 Availability Zone: Asia Pacific (Mumbai) / ap-south-1a
 IPv4 VPC CIDR block: 10.0.0.0/16
 IPv4 subnet CIDR block: 10.0.2.0/24

Tags - optional
 Add new subnet

Create subnet

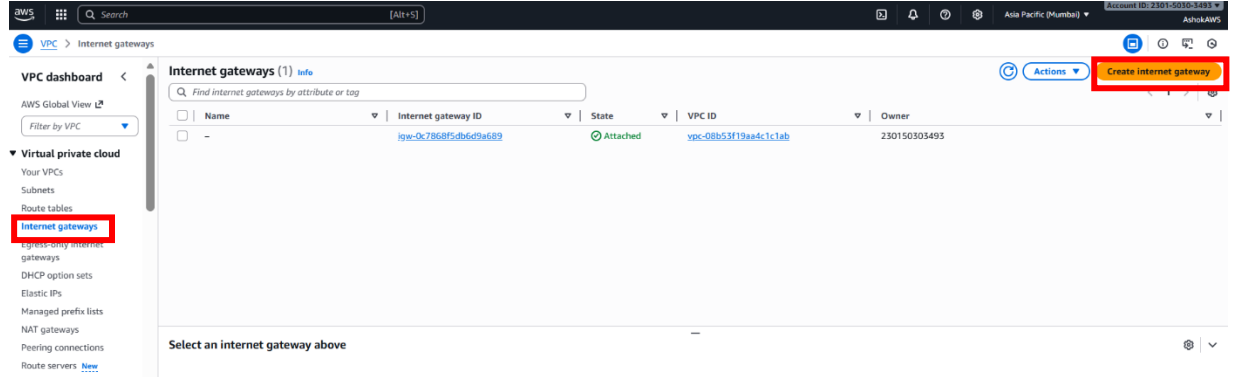
A success message will be displayed after creating subnet (PrivateSubnet).

The screenshot shows the AWS VPC console with the 'Subnets' page. A green success message is displayed at the top: 'You have successfully created 1 subnet: subnet-0930e2df763d2de2f'. The 'Create subnet' button is highlighted in the top right corner.

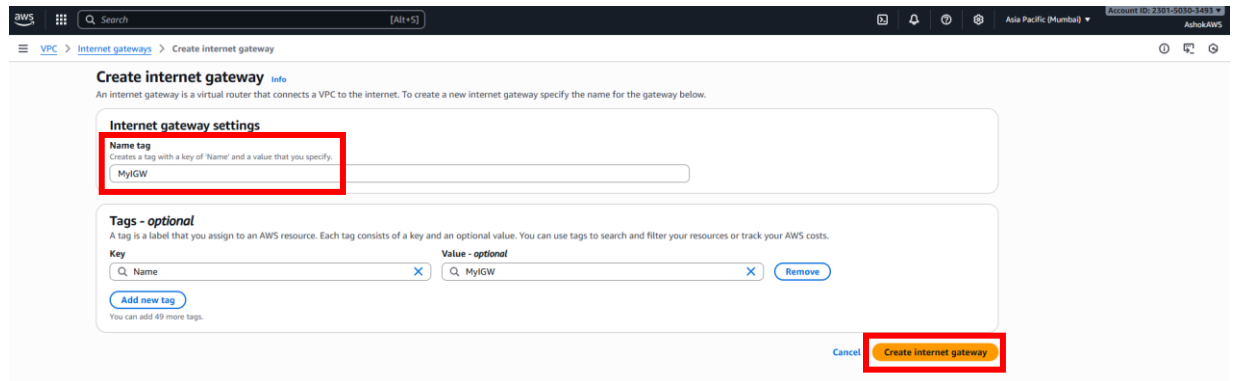
Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 C...
PrivateSubnet	subnet-0930e2df763d2de2f	Available	vpc-078ce8a2d415442a7 MyV...	Off	10.0.2.0/24	-	-

4 Create an Internet Gateway

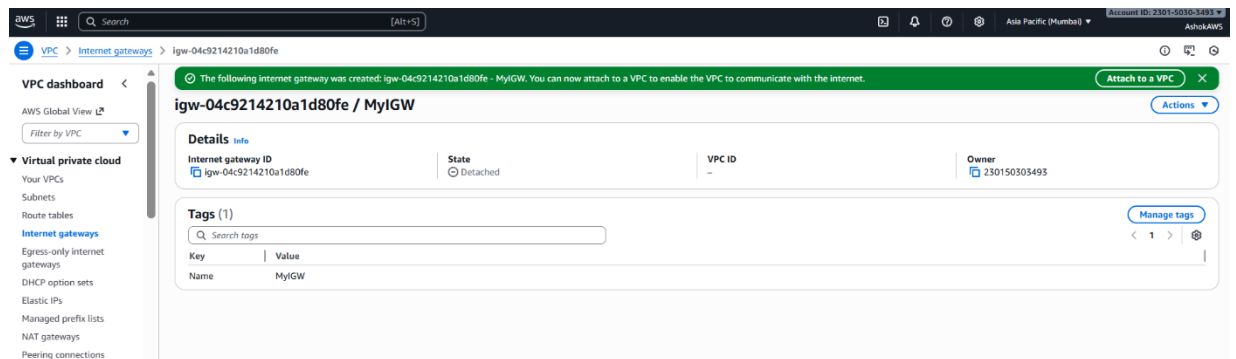
Step 4.1 Click VPC → Internet gateways → Create internet gateway



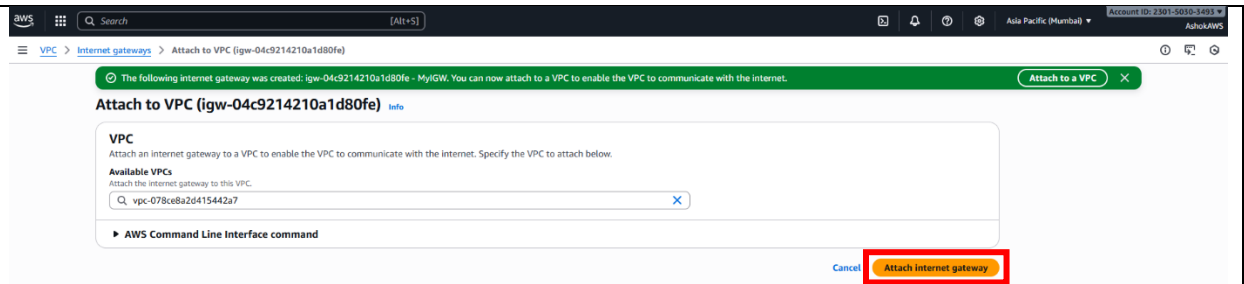
Add the detail shown below, then click “Create internet gateway” button



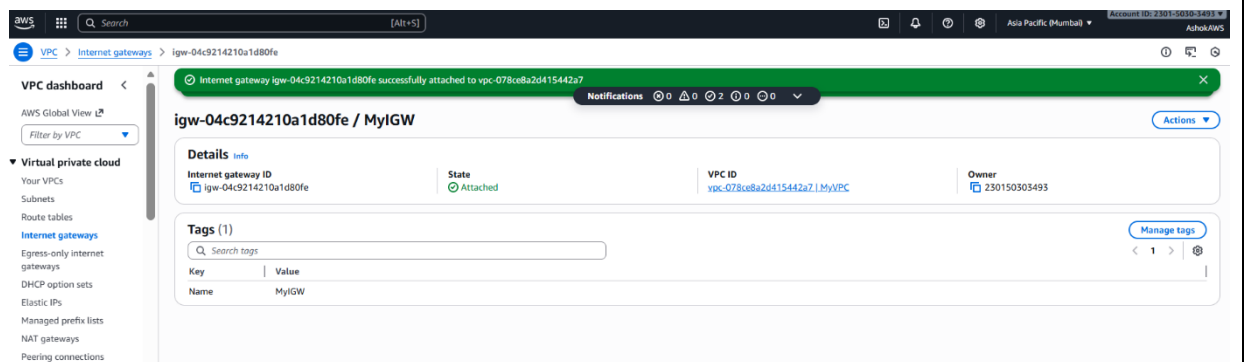
A message will be displayed after creating internet gateway.



Step 4.2 Once the internet gateway is created, attach it to the VPC using the “Attach internet gateway” button.



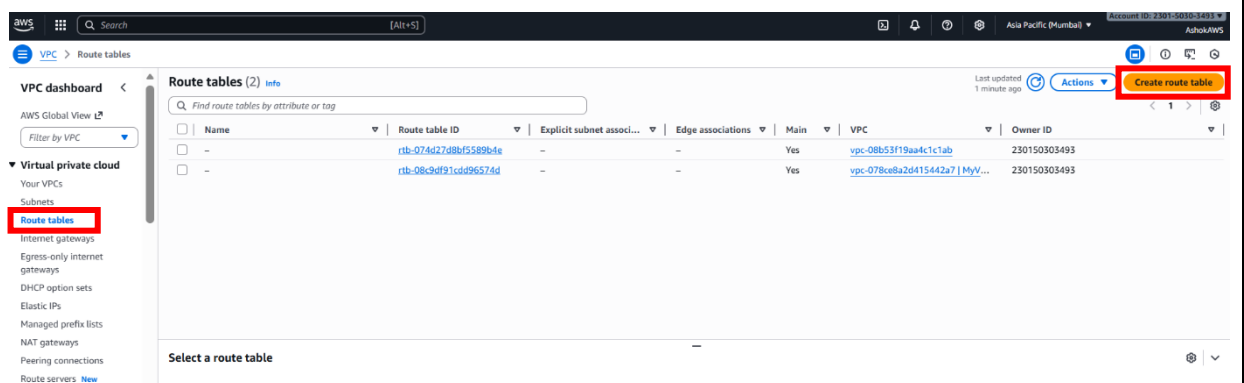
A success message will be displayed after the internet gateway is attached to the VPC.



5 Configure Route Tables

Create Public Route Table

Step 5.1 Click VPC → Route tables → Create route table



Add the details shown below, then click “Create route table” button

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

PublicRouteTable

VPC
The VPC to use for this route table.

vpc-078ce8a2d415442a7 (MyVPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name

Value - optional
Q PublicRouteTable

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create route table](#)

A success message will be displayed after creating route table (PublicRouteTable).

Route table rtb-05cc21258c9875a6f / PublicRouteTable Actions

Details Info

Route table ID: rtb-05cc21258c9875a6f
VPC: vpc-078ce8a2d415442a7 | MyVPC
Main: No
Owner ID: 230150303493

Explicit subnet associations: -
Edge associations: -

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (1) Both [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

Step 5.2 In Routes section, click “Edit routes” button, then click “Add route” button

Add the details:

- **Destination: 0.0.0.0/0**
- **Target: select your Internet Gateway**

then click “Save changes” button

Associate the route table with the public subnet.

Route table rtb-05cc21258c9875a6f / PublicRouteTable Actions

Details Info

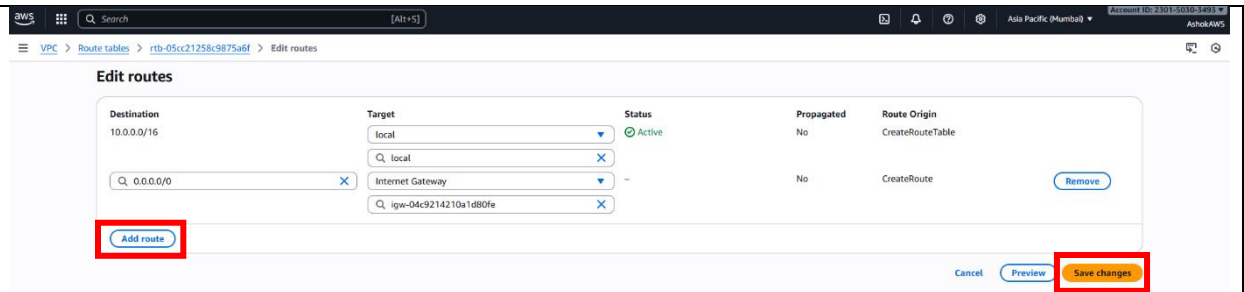
Route table ID: rtb-05cc21258c9875a6f
VPC: vpc-078ce8a2d415442a7 | MyVPC
Main: No
Owner ID: 230150303493

Explicit subnet associations: -
Edge associations: -

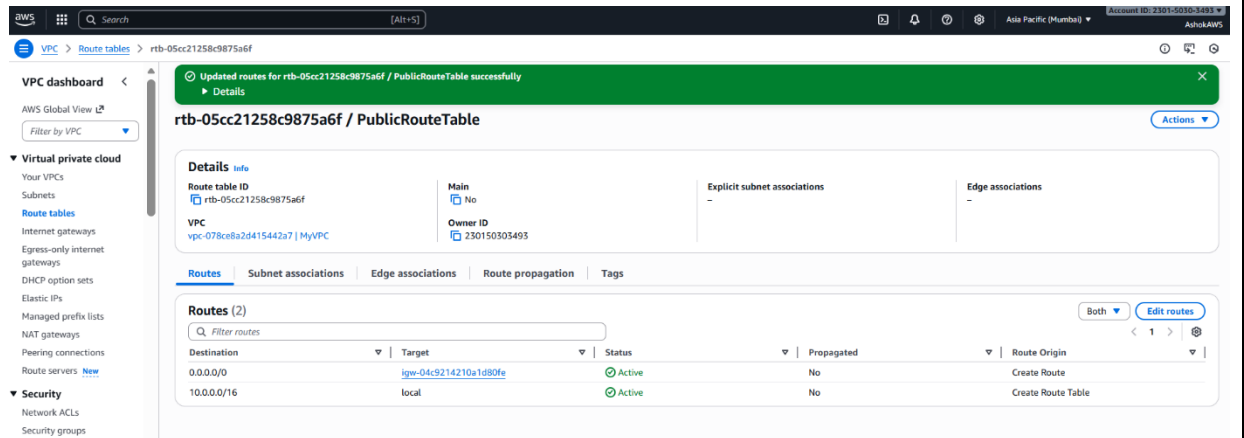
Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (1) Both [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

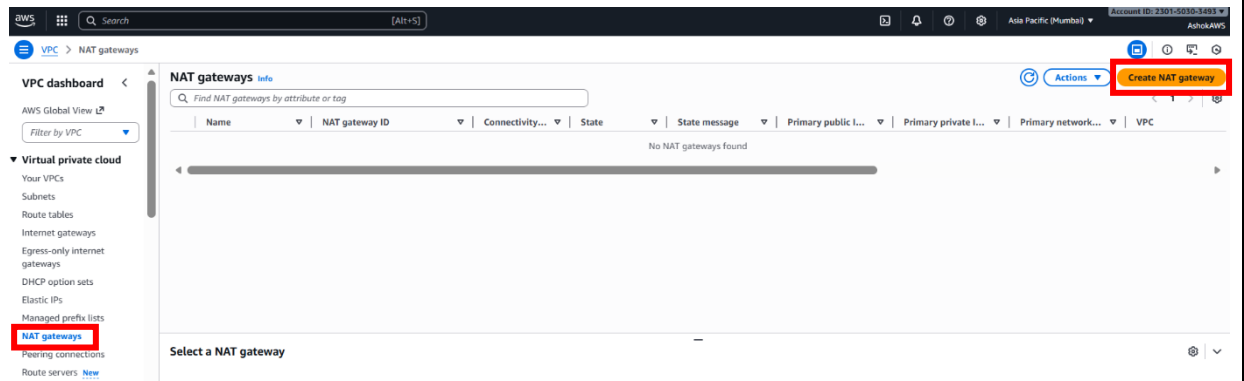


A success message will be displayed after the route is updated for the public route table.



Create Private Route Table

Step 5.3 Click VPC → NAT gateways → Create NAT gateway



Add the details:

- **Subnet:** select your public subnet
- **Elastic IP allocation ID:** Click “Allocate Elastic IP”

then click “Create NAT Gateway” button

Create NAT gateway

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
MyNATGateway

Subnet
Select a subnet in which to create the NAT gateway.
subnet-00bea82fad828183c (PublicSubnet)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID - info
Assign an Elastic IP address to the NAT gateway.
elipalloc-059672e5ca4b0f [Allocate Elastic IP](#)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key - optional
MyNATGateway

Value - optional
MyNATGateway

[Add new tag](#)

[Cancel](#) [Create NAT gateway](#)

A success message will be displayed after creating NAT gateway.

NAT gateway nat-0a14deeab565ff30b | MyNATGateway was created successfully.

nat-0a14deeab565ff30b / MyNATGateway

Details

NAT gateway ID
nat-0a14deeab565ff30b

NAT gateway ARN
arn:aws:ec2:ap-south-1:230150303493:natgateway/nat-0a14deeab565ff30b

VPC
vpc-078ce8a2d415442a7 / MyVPC

Connectivity type
Public

Primary public IPv4 address
-

Subnet
subnet-00bea82fad828183c / PublicSubnet

State
Pending

Primary private IPv4 address
10.0.1.176

Created
Wednesday, November 5, 2025 at 17:22:43 GMT+5:30

State message - info
-

Primary network interface ID
eni-09304827a50764885 [L](#)

Deleted
-

Secondary IPv4 addresses

[Edit secondary IPv4 address associations](#)

Private IPv4 address | Network interface ID | Status | Failure message

Secondary IPv4 addresses are not available for this nat gateway.

Update the Private Route Table

Now, configure the private subnet to use the NAT Gateway for internet traffic.

Step 5.4 Click VPC → Route tables → Create route table

Route tables (3)

[Find route tables by attribute or tag](#)

[Actions](#) [Create route table](#)

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-074427d8b45589b4e	-	-	Yes	vpc-08b53f19ap4c1c1ab	230150303493
PublicRouteTable	rtb-05c21258c9875a6f	-	-	No	vpc-078ce8a2d415442a7 MyV...	230150303493
-	rtb-08c9df91cd996574d	-	-	Yes	vpc-078ce8a2d415442a7 MyV...	230150303493

Select a route table

Step 5.5 If you don't have a PrivateRouteTable yet, create one:
Add the details:

- **Name:** PrivateRouteTable
- **VPC:** select your VPC
- Click **Create route table**

Create route table [info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

PrivateRouteTable

VPC
The VPC to use for this route table.

vpc-078ce8a2d415442a7 (MyVPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name

Value - optional
Q PrivateRouteTable

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create route table](#)

A success message will be displayed after creating route table (PrivateRouteTable).

Route table rtb-0ef867a3ad61b101f | PrivateRouteTable was created successfully.

rtb-0ef867a3ad61b101f / PrivateRouteTable [Actions](#)

Details [info](#)

Route table ID: rtb-0ef867a3ad61b101f

VPC: vpc-078ce8a2d415442a7 | MyVPC

Main: No

Owner ID: 230150303493

Explicit subnet associations: -

Edge associations: -

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (1)

[Filter routes](#)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

Step 5.6 Click **Route tables** → go to **Routes** tab → click **Edit routes** → click **Add route**
Add the details:

- **Destination:** 0.0.0.0/0
- **Target:** your new NAT Gateway (nat-xxxxxxx)
- Click **Save changes**

Route table rtb-0ef867a3ad61b101f / PrivateRouteTable

Details info

Route table ID: rtb-0ef867a3ad61b101f

VPC: vpc-078ce8a2d415442a7 | MyVPC

Main: No

Owner ID: 230150303493

Explicit subnet associations: -

Edge associations: -

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (1)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

Edit routes

Destination: 10.0.0.0/16

Target: local

Status: Active

Propagated: No

Route Origin: CreateRouteTable

Q: 0.0.0.0/0

NAT Gateway

nat-0e14deeaab565ff30b

Add route

Cancel | Preview | **Save changes**

A success message will be displayed after the route is updated for the private route table.

Route table rtb-0ef867a3ad61b101f / PrivateRouteTable

Updated routes for rtb-0ef867a3ad61b101f / PrivateRouteTable successfully

Details

Route table ID: rtb-0ef867a3ad61b101f

VPC: vpc-078ce8a2d415442a7 | MyVPC

Main: No

Owner ID: 230150303493

Explicit subnet associations: -

Edge associations: -

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0e14deeaab565ff30b	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Step 5.7 Go to **Subnet associations** tab → click “**Edit subnet associations**” button

Subnet associations

Explicit subnet associations (0)

Subnets without explicit associations (2)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
PublicSubnet	subnet-00bea82fad828183c	10.0.1.0/24	-
PrivateSubnet	subnet-0930e2df763d2de2f	10.0.2.0/24	-

Select Private Subnet and click **“Save associations”** button

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
PublicSubnet	subnet-00bea82fad828183c	10.0.1.0/24	-	Main (rtb-08c9d)
☒ PrivateSubnet	subnet-0930e2df763d2de2f	10.0.2.0/24	-	Main (rtb-08c9d)

Selected subnets

subnet-0930e2df763d2de2f / PrivateSubnet

Save associations

A success message will be displayed after the subnet associations is updated for the private route table.

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0a140eeab565f930b	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Thus, the VPC has been created successfully with public and private subnets.

